Cheatsheet · 15 chapters on one page

Maths for Machine Learning cheatsheet

Every formula, key idea, table, memory aid and common mistake from the chapters, gathered in one place for quick revision. Nothing here is new. Each card links back to the lesson that teaches it, so open the lesson when a line does not make sense yet.

105 key ideas | 210 formulas | 79 tables | 80 tips and traps

STEP 1 School foundations

1.1 Numbers, Algebra and Equations

The school maths underneath every model: kinds of numbers, the order of operations, percentages, equations that stay balanced, powers, the sigma sign and absolute value. Each idea is linked to how a machine learning model uses it.

KEY IDEAS

- **Why machine learning needs maths** A machine learning model is a function with knobs . You give it numbers (the size of a house, the pixels of a photo, the words of a message) and it gives back a number (a price, the word "cat", the chance that a message is spam).
- **Kinds of numbers and how computers store them** Numbers come in families. Each family is bigger than the one before and contains it.
- **Order of operations, fractions and percentages** When an expression mixes operations, everyone must work it out in the same order, or two people would get two different answers.
- **Variables, expressions and equations** A variable is a box with a name, such as x, that holds a number we do not know yet. An expression such as 3x + 5 is a recipe that uses the box. An equation such as 3x + 5 = 20 says two things are equal, and solving it means finding what is in the box.
- **Powers, roots and scientific notation** A power is repeated multiplication: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$. The small number 5 is the exponent and 2 is the base .
- **Sigma and Pi notation: the maths shorthand** Machine learning formulas add up many terms at once. Writing $x_1 + x_2 + x_3 + \ldots + x_n$ is slow, so mathematicians use the Greek capital letter sigma, Σ, which means "sum".
- **Absolute value, inequalities and distance** The absolute value |x| is the distance of x from zero, so it is never negative: |5| = 5 and |−5| = 5. The distance between two numbers a and b is |a − b|, whichever is bigger.

FORMULAS

Machine epsilon Exam favourite

$$\varepsilon = 2^{-52} \approx 2.2 \times 10^{-16} \text{ for float64}$$

The gap between 1 and the next number a double can hold. Any relative error smaller than this is invisible to the computer.

Lesson: Kinds of numbers and how computers store them

BODMAS (also called PEMDAS) Exam favourite

$$\text{B} \rightarrow \text{O} \rightarrow \text{D·M} \rightarrow \text{A·S}$$

Division and multiplication have equal rank, so go left to right. The same is true for addition and subtraction.

Lesson: Order of operations, fractions and percentages

Percentage

$$\text{percent} = \frac{\text{part}}{\text{whole}} \times 100$$

Accuracy of 46 correct out of 50 is 46 ÷ 50 × 100 = 92%.

Lesson: Order of operations, fractions and percentages

Percentage change

$$\text{change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

Going from 80% to 92% accuracy is a 15% improvement, not 12%.

Lesson: Order of operations, fractions and percentages

Two equations, two unknowns (Cramer rule)

Exam favourite

$$a_1x + b_1y = c_1 \text{ and } a_2x + b_2y = c_2 \text{ give } x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1} \text{ and } y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

The bottom part $a_1b_2 - a_2b_1$ is called the determinant. If it is 0, the two lines are parallel and there is no single answer. Chapter 4 meets it again.

Lesson: Variables, expressions and equations

Sigma means add them all

Exam favourite

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$$

i is a counter that starts at the number below and stops at the number on top. Read it as "the sum of x i, for i from 1 to n".

Lesson: Sigma and Pi notation: the maths shorthand

Mean (average)

$$\bar{x} = \frac{1}{n} \sum x_i$$

Add everything and divide by how many there are. The bar on top of x is read "x bar".

Lesson: Sigma and Pi notation: the maths shorthand

Capital pi means multiply them all

$$\prod x_i = x_1 \times x_2 \times \ldots \times x_n$$

Probabilities of independent events multiply, so Π appears in likelihoods (Chapter 11).

Lesson: Sigma and Pi notation: the maths shorthand

Absolute value

Exam favourite

$$|x| = x \text{ if } x \geq 0, \text{ and } -x \text{ if } x < 0$$

It removes the minus sign. $|3 - 7| = |-4| = 4$.

Lesson: Absolute value, inequalities and distance

Mean absolute error

$$\text{MAE} = \frac{1}{n} \sum |y_i - \hat{y}_i|$$

y is the true value and ŷ (read "y hat") is the prediction. Average distance between them.

Lesson: Absolute value, inequalities and distance

Triangle inequality

$$|a + b| \leq |a| + |b| \quad \text{and} \quad |a - c| \leq |a - b| + |b - c|$$

Going from a to c directly is never longer than going through b.

Lesson: Absolute value, inequalities and distance

TABLES

Table: family · symbol · examples · everyday use

Family	Symbol	Examples	Everyday use
Natural numbers	ℕ	1, 2, 3, 4	Counting apples, number of rows in a dataset
Whole numbers		0, 1, 2, 3	Number of goals, zero included
Integers	ℤ	−3, −1, 0, 2	Temperature below zero, a bank balance
Rational numbers	ℚ	½, 0.75, −4/3	Any value you can write as one whole number over another
Irrational numbers		√2, π, e	Decimals that never end and never repeat
Real numbers	ℝ	all of the above	Every point on the number line. Model inputs live here

Lesson: Kinds of numbers and how computers store them

Table: format · sign · exponent bits · fraction bits

Format	Sign	Exponent bits	Fraction bits	Correct decimal digits	Where you meet it
float64 (double)	1	11	52	about 15 to 16	Python float, JavaScript number, Java double
float32 (single)	1	8	23	about 7	the default in PyTorch and on most GPUs
float16 (half)	1	5	10	about 3	mixed precision training, largest value only 65504
bfloat16	1	8	7	about 2 to 3	Google TPUs and modern GPUs, same range as float32

Lesson: Kinds of numbers and how computers store them

Table: phrase · what it measures · example: accuracy goes from 80% to 92%

Phrase	What it measures	Example: accuracy goes from 80% to 92%
percentage points	the plain difference between two percentages	92 − 80 = 12 percentage points
percent change	the difference as a share of the starting value	12 ÷ 80 × 100 = 15% better
error rate change	often the most honest view for good models	errors fell from 20% to 8%, a 60% cut in mistakes

Lesson: Order of operations, fractions and percentages

Table: rule · in symbols · example

Rule	In symbols	Example
Multiply, add the exponents	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7 = 128$
Divide, subtract the exponents	$a^m \div a^n = a^{m-n}$	$2^{10} \div 2^7 = 2^3 = 8$
Power of a power, multiply	$(a^m)^n = a^{mn}$	$(2^3)^2 = 2^6 = 64$
Anything to the power 0	$a^0 = 1$	$5^0 = 1$
Negative exponent, flip it	$a^{-n} = \frac{1}{a^n}$	$2^{-3} = \frac{1}{8} = 0.125$
Fractional exponent, a root	a^(1/n) = $\sqrt[n]{a}$	9^(1/2) = √9 = 3

Lesson: Powers, roots and scientific notation

Table: power · 2^4 · 2^3 · 2^2

Power	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}
Value	16	8	4	2	1	½	¼
How you got there		÷ 2	÷ 2	÷ 2	÷ 2	÷ 2	÷ 2

Lesson: Powers, roots and scientific notation

Table: rule · in symbols · why it is true

Rule	In symbols	Why it is true
Split a sum	$\Sigma (a_i + b_i) = \Sigma a_i + \Sigma b_i$	you can add the numbers in any order
Pull out a constant	$\Sigma c{\cdot}a_i = c \cdot \Sigma a_i$	$c(a_1 + a_2) = ca_1 + ca_2$, the distributive law
Sum of a constant	Σ from 1 to n of c = n · c	adding c to itself n times
Sum of a log	$\Sigma \log a_i = \log \Pi a_i$	logs turn products into sums (Chapter 2)

Lesson: Sigma and Pi notation: the maths shorthand

Table: written · meaning · on the number line

Written	Meaning	On the number line
$\|x\| < 3$	x is closer than 3 to zero	between −3 and 3, written (−3, 3)
$\|x - 10\| \le 2$	x is within 2 of 10	from 8 to 10 inclusive, written [8, 12]
$x \ge 0$	x is not negative	zero and everything to its right

Lesson: Absolute value, inequalities and distance

TIPS AND TRAPS

COMMON MISTAKE Never test two decimal results with ==. Test whether they are close: `abs(a - b) < 1e-9`. NumPy calls this `np.isclose`, and PyTorch calls it `torch.allclose`.

Lesson: Kinds of numbers and how computers store them

REMEMBER Percent means "per hundred". To find 15% of anything, find 10% (move the decimal point once) and add half of it. 15% of 240 is 24 + 12 = 36.

Lesson: Order of operations, fractions and percentages

REMEMBER Think of "undoing" the steps in reverse order, like taking off socks and shoes. The equation $3x + 5 = 20$ was built by "times 3, then plus 5". So undo it with "minus 5, then divide by 3".

Lesson: Variables, expressions and equations

REMEMBER Undo the operations in the reverse order they were done, like unwrapping a gift: the last layer of paper comes off first.

Lesson: Variables, expressions and equations

REMEMBER The exponent of 10 tells you how many places to move the decimal point. Positive moves right (bigger), negative moves left (smaller). 2.5e3 = 2500 and 2.5e-3 = 0.0025.

Lesson: Powers, roots and scientific notation

REMEMBER Σ is a for loop. The counter under the Σ is the loop variable, the numbers under and over it are where the loop starts and stops, and the expression after it is the loop body that gets added.

Lesson: Sigma and Pi notation: the maths shorthand

REMEMBER Gauss as a schoolboy added 1 + 2 + ... + 100 by pairing the first and last numbers: 1 + 100, 2 + 99 and so on make 50 pairs of 101, so the total is 5050. In general Σ i from 1 to n = n(n + 1) ÷ 2.

Lesson: Sigma and Pi notation: the maths shorthand

IN THE EXAM $|x - 10| \le 2$ means $8 \le x \le 12$, not $8 \le x \le 10$. Students often forget the right half of the interval.

Lesson: Absolute value, inequalities and distance

1.2 Functions, Graphs, Exponents and Logarithms

Functions are the machines models are made of. Learn straight lines, parabolas, exponential growth, the number e, logarithms and the sigmoid, ReLU and tanh activations, with sliders and graphs you can drag.

KEY IDEAS

- **What a function is** A function is a machine with one rule: every input gives exactly one output. A vending machine is a function. Press B4 and you always get the same snack. We write the machine as f and its output for input x as f(x), read "f of x".
- **Straight lines: slope and intercept** The simplest useful function is a straight line. Two numbers describe it completely.
- **Quadratics and polynomials** A quadratic has an x^2 term: $f(x) = ax^2 + bx + c$. Its graph is a parabola , a U shape when $a > 0$ and an upside down U when $a < 0$. The bottom (or top) is called the vertex .

- **Exponential growth and the number e** In an exponential function the variable sits in the exponent: $f(x) = a^x$. Each step multiplies instead of adding. With $a > 1$ it grows faster and faster, with $0 < a < 1$ it shrinks towards zero but never reaches it.
- **Logarithms: the undo button for powers** A logarithm answers the question "what power do I need?". Since $2^3 = 8$, $\log_2 8 = 3$. Since $10^2 = 100$, $\log_{10} 100 = 2$. The natural log, ln, uses base e.
- **The S curve and other functions machine learning uses** Some functions show up in machine learning again and again. They are called activation functions because they decide how strongly a neuron "fires".
- **Composing, shifting and inverting functions** Composition feeds one function into another: $(f \circ g)(x) = f(g(x))$. Work from the inside out. If $g(x) = 3x - 1$ and f is the sigmoid, then $f(g(1)) = \sigma(2) \approx 0.88$.

FORMULAS

Function notation

Exam favourite

$$f(x) = 2x + 1 \implies f(3) = 2 \times 3 + 1 = 7$$

Replace every x in the rule with the input, then work it out.

Lesson: What a function is

A piecewise function

$$f(x) = 0 \text{ if } x < 0, \text{ and } f(x) = x \text{ if } x \geq 0$$

This is ReLU. Each piece is simple, and together they make a bend that a neural network can use to build curved shapes.

Lesson: What a function is

Slope intercept form

Exam favourite

$$y = m\,x + b$$

m is the slope (how much y climbs when x goes up by 1) and b is the intercept (the value of y when x = 0).

Lesson: Straight lines: slope and intercept

Slope from two points

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{rise}}{\text{run}}$$

Positive slope climbs, negative slope falls, zero slope is flat.

Lesson: Straight lines: slope and intercept

Quadratic formula

Exam favourite

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

Gives the roots, where the curve crosses y = 0.

Lesson: Quadratics and polynomials

Discriminant

$$D = b^2 - 4ac$$

D > 0: two roots. D = 0: one root. D < 0: no real roots.

Lesson: Quadratics and polynomials

Vertex

$$x = \frac{-b}{2a}$$

The lowest point when a > 0. This is the minimum we look for in learning.

Lesson: Quadratics and polynomials

Vertex form

$$f(x) = a(x - h)^2 + k$$

The vertex is (h, k). If a > 0 the curve opens upward and k is the lowest value. If a < 0 it opens downward and k is the highest.

Lesson: Quadratics and polynomials

Euler number

Exam favourite

$$e = \lim (1 + \frac{1}{n})^n \text{ as } n \to \infty \approx 2.71828$$

Read lim as "the value it settles at when n grows without end".

Lesson: Exponential growth and the number e

Natural growth and decay

$$A(t) = A_0\, e\text{^}(kt)$$

k > 0 grows, k < 0 decays. A_0 is the starting amount.

Lesson: Exponential growth and the number e

Doubling time

$$T_2 = \frac{\ln 2}{k} \approx \frac{0.693}{k}$$

For growth A_0e^{kt}. The "rule of 72" says money at r% a year doubles in about 72 ÷ r years.

Lesson: Exponential growth and the number e

Half life

$$T½ = \frac{\ln 2}{k} \text{ for } A_0e^{-kt}$$

Carbon 14 has a half life of about 5730 years, which is how old bones are dated.

Lesson: Exponential growth and the number e

Definition

Exam favourite

$$\log_b(y) = x \quad \text{means} \quad b^x = y$$

The log is the exponent. ln(x) means log base e, and $\log_2$ counts doublings, or bits.

Lesson: Logarithms: the undo button for powers

Sigmoid

Exam favourite

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Squashes any number into (0, 1). σ(0) = 0.5. Used for probabilities of yes or no.

Lesson: The S curve and other functions machine learning uses

Tanh

$$\tanh(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

Squashes into (−1, 1) and is centred on 0.

Lesson: The S curve and other functions machine learning uses

ReLU

$$\mathrm{ReLU}(z) = \max(0, z)$$

Zero for negatives, unchanged for positives. Fast, and the default in deep networks.

Lesson: The S curve and other functions machine learning uses

Composition

Exam favourite

$$(f \circ g)(x) = f(g(x))$$

Order matters: f(g(x)) and g(f(x)) are usually different.

Lesson: Composing, shifting and inverting functions

Logit, the inverse of sigmoid

$$\mathrm{logit}(p) = \ln \frac{p}{1 - p}$$

p ÷ (1 − p) is called the odds. Logit turns a probability back into a score.

Lesson: Composing, shifting and inverting functions

TABLES

Table: word · meaning · example · why ml cares

Word	Meaning	Example	Why ML cares
increasing	a bigger input always gives a bigger (or equal) output	e^x, the sigmoid, ln x	the order of scores is kept, so the biggest logit is still the biggest probability
one to one	no two inputs share an output	2x + 1 yes, x^2 no (since $3^2 = (-3)^2 = 9$)	only one to one functions can be undone exactly, which is why logit can undo sigmoid
bounded	outputs stay between two fixed numbers	sigmoid between 0 and 1	good for probabilities, but flat ends slow down learning

Lesson: What a function is

Table: form · formula · use it when you know

Form	Formula	Use it when you know
slope intercept	$y = mx + b$	the slope and where it crosses the y axis
point slope	$y - y_1 = m(x - x_1)$	the slope and any one point on the line
two point	$m = (y_2 - y_1) \div (x_2 - x_1)$, then point slope	two points on the line

Lesson: Straight lines: slope and intercept

Table: x · 0 · 1 · 2

x	0	1	2	3	4	10
2x (adding)	0	2	4	6	8	20
2^x (multiplying)	1	2	4	8	16	1024
0.5^x (halving)	1	0.5	0.25	0.125	0.0625	0.00098

Lesson: Exponential growth and the number e

Table: rule · in symbols · example

Rule	In symbols	Example
Product becomes sum	$\log(ab) = \log a + \log b$	$\log_2(4 \times 8) = 2 + 3 = 5$
Quotient becomes difference	$\log(a \div b) = \log a - \log b$	$\log_{10}(1000 \div 10) = 3 - 1 = 2$
Power comes down in front	$\log(a^n) = n \log a$	$\ln(e^5) = 5$
Log of 1 is 0	$\log 1 = 0$	because $b^0 = 1$
Change of base	$\log_b x = \frac{\ln x}{\ln b}$	$\log_2 100 = \frac{4.605}{0.693} \approx 6.64$

Lesson: Logarithms: the undo button for powers

Table: probability p · ln p · reading

Probability p	ln p	Reading
1	0	certain: no penalty
0.5	−0.693	a coin flip
0.1	−2.303	unlikely
0.001	−6.908	very unlikely
→ 0	→ −∞	impossible: an infinite penalty

Lesson: Logarithms: the undo button for powers

Table: function · output range · good for · weak spot

Function	Output range	Good for	Weak spot
Sigmoid	0 to 1	the last layer of a yes or no classifier	flat at both ends, so learning slows down
Tanh	−1 to 1	hidden layers in older networks and RNNs	also flat at both ends
ReLU	0 to ∞	hidden layers of almost every modern network	a neuron stuck below 0 stops learning
Softmax	0 to 1, adding to 1	choosing one class out of many	needs care with large numbers

Lesson: The S curve and other functions machine learning uses

Table: activation · formula · where it is used

Activation	Formula	Where it is used
Leaky ReLU	$\max(0.01x, x)$	keeps a small slope for negatives so neurons never fully die
Softplus	$\ln(1 + e^x)$	a smooth ReLU whose slope is the sigmoid
GELU	$x \cdot \Phi(x)$, Φ is the normal CDF	BERT, GPT style transformers
SiLU (Swish)	$x \cdot \sigma(x)$	many modern networks, and inside the SwiGLU blocks of several large language models

Lesson: The S curve and other functions machine learning uses

Table: change to f(x) · what the graph does

Change to f(x)	What the graph does
f(x) + k	moves up by k
f(x − h)	moves right by h
a · f(x)	stretches up and down by a, and flips if a < 0
f(c · x)	squeezes sideways by c

Lesson: Composing, shifting and inverting functions

TIPS AND TRAPS

REMEMBER Slope is "rise over run", like a staircase: how far up for each step forward. Parallel lines have equal slopes. Perpendicular lines have slopes that multiply to −1.

Lesson: Straight lines: slope and intercept

REMEMBER "Logs turn multiplying into adding". That single sentence explains slide rules, decibels, the Richter scale and why every ML loss for probabilities has a log in it.

Lesson: Logarithms: the undo button for powers

REMEMBER Log of a number between 0 and 1 is always negative. That is why losses are written as MINUS log probability, so that the loss is a positive number and smaller is better.

Lesson: Logarithms: the undo button for powers

IN THE EXAM ln(0) does not exist and ln of a negative number is not real. Code adds a tiny epsilon, as in log(p + 1e-12), so a predicted probability of exactly 0 does not crash training.

Lesson: Logarithms: the undo button for powers

REMEMBER The graph of an inverse function is the mirror image of the original in the line y = x. Swap the x and y of every point and you have the inverse.

Lesson: Composing, shifting and inverting functions

STEP 2 Linear algebra

2.1 Vectors

Vectors are lists of numbers and arrows at the same time. Add and scale them, measure their length, compare them with the dot product and cosine similarity, cast shadows with projections, and learn what span, independence and a basis mean.

KEY IDEAS

- **What a vector is** A vector is an ordered list of numbers. You can picture it in two ways, and both are useful.
- **Adding, subtracting and scaling vectors** Vectors are added part by part: [3, 1] + [1, 2] = [4, 3]. As arrows, put the tail of the second at the tip of the first. The sum goes from the very start to the very end. Multiplying by a number (a scalar) stretches the arrow: 2 × [3, 1] = [6, 2].
- **Length (norm) and distance** The length of a vector is called its norm . The usual one comes straight from Pythagoras: [3, 4] has length $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$. There are other useful ways to measure size.
- **The dot product** The dot product turns two vectors into one number. Multiply matching parts, then add: [2, 3] · [4, −1] = 2 × 4 + 3 × (−1) = 8 − 3 = 5. It also has a meaning in pictures: it measures how much two arrows point the same way.
- **Cosine similarity: how alike are two things** If you divide the dot product by both lengths, you get cos θ alone. This is cosine similarity .
- **Projection: the shadow of a vector** Shine a light straight down onto a line. The shadow a vector casts on that line is its projection . It tells you how much of one vector lies along the direction of another.

- **Span, independence and basis** Take all the linear combinations of some vectors. Every point you can reach is called their span . With [1, 0] and [0, 1] you can reach any point in the plane: [5, −2] = 5·[1, 0] − 2·[0, 1].

FORMULAS

Writing a vector

Exam favourite

$$v = [v_1, v_2, \ldots, v_n] \quad \text{or as a column} \quad \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$$

Books print vectors in bold (**v**) or with an arrow on top. $\mathbb{R}^n$ means "all vectors with n real numbers".

Lesson: What a vector is

Addition

Exam favourite

$$a + b = [a_1 + b_1, a_2 + b_2, \ldots]$$

Only for vectors of the same size.

Lesson: Adding, subtracting and scaling vectors

Scalar multiplication

$$k \cdot a = [k \cdot a_1, k \cdot a_2, \ldots]$$

k > 1 stretches, 0 < k < 1 shrinks, k < 0 flips.

Lesson: Adding, subtracting and scaling vectors

Linear combination

$$c_1 \cdot v_1 + c_2 \cdot v_2 + \ldots + c_k \cdot v_k$$

Scale several vectors, then add them. This is the most important pattern in linear algebra.

Lesson: Adding, subtracting and scaling vectors

Centroid (average vector)

$$c = \frac{1}{n}(v_1 + v_2 + \ldots + v_n)$$

Add the vectors, then scale by 1 ÷ n. The result sits in the middle of the cloud of points.

Lesson: Adding, subtracting and scaling vectors

Interpolation (lerp)

$$v(t) = (1 - t)\, a + t\, b, \quad 0 \le t \le 1$$

t = 0 gives a, t = 1 gives b, t = ½ gives the halfway point. Every point in between lies on the straight segment from a to b.

Lesson: Adding, subtracting and scaling vectors

L2 norm (Euclidean)

Exam favourite

$$\|v\|_2 = \sqrt{(v_1^2 + v_2^2 + \ldots + v_n^2)}$$

The straight line, as the crow flies.

Lesson: Length (norm) and distance

L1 norm (Manhattan)

$$\|v\|_1 = |v_1| + |v_2| + \ldots + |v_n|$$

Walking along city blocks.

Lesson: Length (norm) and distance

Unit vector

$$\hat{u} = \frac{v}{\|v\|}$$

Same direction, length exactly 1. This is called normalising.

Lesson: Length (norm) and distance

Distance between points

$$d(p, q) = \|p - q\|$$

Subtract, then take the length.

Lesson: Length (norm) and distance

Lp norm

$$\|v\|_p = (|v_1|^p + |v_2|^p + \ldots + |v_n|^p)\text{^}(1/p)$$

p = 1 gives Manhattan, p = 2 gives Euclidean, and as p grows huge the biggest component wins, giving $L\infty$ = max $|v_i|$.

Lesson: Length (norm) and distance

Algebra form

Exam favourite

$$a \cdot b = a_1 b_1 + a_2 b_2 + \ldots + a_n b_n = \Sigma\, a_i b_i$$

Easy to compute, even in a million dimensions.

Lesson: The dot product

Geometry form

$$a \cdot b = \|a\| \, \|b\| \cos \theta$$

θ (theta) is the angle between the arrows.

Lesson: The dot product

Cosine similarity

Exam favourite

$$\cos(a, b) = \frac{a \cdot b}{\|a\| \, \|b\|}$$

1 means same direction, 0 means unrelated, −1 means opposite. For counts, which are never negative, it stays between 0 and 1.

Lesson: Cosine similarity: how alike are two things

Cosine distance

$$1 - \cos(a, b)$$

0 for the same direction, 1 for perpendicular, 2 for opposite.

Lesson: Cosine similarity: how alike are two things

Scalar projection (shadow length)

Exam favourite

$$\text{comp_b a} = \frac{a \cdot b}{\|b\|}$$

A signed length. Negative means the shadow points backwards.

Lesson: Projection: the shadow of a vector

Vector projection (the shadow itself)

$$\text{proj_b a} = \frac{a \cdot b}{b \cdot b} \, b$$

A vector along b. What is left over, a – proj_b a, is perpendicular to b.

Lesson: Projection: the shadow of a vector

Decomposition

$$a = \text{proj_b a} + (a - \text{proj_b a})$$

The first piece is parallel to b. The second piece is perpendicular to b, so its dot product with b is 0.

Lesson: Projection: the shadow of a vector

Basis and dimension

Exam favourite

basis = a set of independent vectors that spans the space

Every basis of $\mathbb{R}^n$ has exactly n vectors. That number n is the dimension.

Lesson: Span, independence and basis

TABLES

Table: kind of vector · what it looks like · example

Kind of vector	What it looks like	Example
feature vector	one number per measured property	[area 85, rooms 2, age 12] for one flat
one hot vector	all zeros except a single 1	colour "green" out of [red, green, blue] is [0, 1, 0]
embedding	a few hundred learned numbers, no single one has an obvious meaning	a 768 number vector for the word "river"
probability vector	non negative numbers that add up to 1	a classifier output [0.7, 0.2, 0.1]
weight vector	one learned number per input	the slopes of a linear model

Lesson: What a vector is

Table: norm · value for [3, −4] · ml use

Norm	Value for [3, −4]	ML use
L2	5	k nearest neighbours, k means, Ridge (L2) regularisation
L1	7	Lasso (L1) regularisation, which pushes weights to exactly zero
L∞	4	adversarial examples that change each pixel by at most a tiny amount

Lesson: Length (norm) and distance

Table: angle θ · cos θ · dot product · meaning

Angle θ	cos θ	Dot product	Meaning
0°	1	largest positive	same direction
less than 90°	between 0 and 1	positive	roughly the same way
90°	0	zero	**orthogonal** (perpendicular): unrelated
more than 90°	negative	negative	roughly opposite
180°	−1	most negative	exactly opposite

Lesson: The dot product

Table: words · plain meaning

Words	Plain meaning
Span	everywhere you can reach by scaling and adding
Independent	no vector is wasted
Basis	the smallest set of building blocks that reaches everything
Dimension	how many building blocks a basis needs
Orthonormal basis	building blocks of length 1 that are all perpendicular, such as [1, 0] and [0, 1]

Lesson: Span, independence and basis

TIPS AND TRAPS

REMEMBER Vector rules are the same as number rules you already know: a + b = b + a, (a + b) + c = a + (b + c), and k(a + b) = ka + kb. If a step would be fine with ordinary numbers, it is almost always fine with vectors of the same size.

Lesson: Adding, subtracting and scaling vectors

REMEMBER A 3, 4, 5 triangle is the easiest Pythagoras check to remember. Its multiples 6, 8, 10 and 30, 40, 50 work too, and so do 5, 12, 13 and 8, 15, 17.

Lesson: Length (norm) and distance

COMMON MISTAKE Distances are only fair when features use similar scales. If one feature is salary in rupees (around 500000) and another is age in years (around 30), salary dominates every distance. Scale the features first, for example to a mean of 0 and a standard deviation of 1.

Lesson: Length (norm) and distance

REMEMBER Cosine similarity is the dot product of the two unit vectors. Normalise first, then a plain dot product gives the cosine. Vector databases store normalised vectors for exactly this reason.

Lesson: Cosine similarity: how alike are two things

2.2 Matrices and Matrix Multiplication

Matrices hold datasets, images and the weights of every layer. Multiply them step by step with an animation you can edit, watch a matrix bend the plane, find determinants and inverses, solve systems with elimination, and fit a regression

line with the normal equations.

KEY IDEAS

- **What a matrix is** A matrix is a rectangle of numbers arranged in rows and columns. Its shape is written rows × columns, so a matrix with 2 rows and 3 columns is 2 × 3. The entry in row i and column j is written a_{ij}.
- **Adding, scaling and transposing** Matrices of the same shape add entry by entry. Multiplying by a number scales every entry. The transpose A^T flips a matrix over its diagonal, so rows become columns and a 2 × 3 matrix becomes 3 × 2.
- **Matrix multiplication, step by step** To multiply A by B, each entry of the answer is a dot product : row i of A with column j of B. Walk along the row and down the column together, multiply the pairs, and add.
- **A matrix is a transformation** Multiplying a vector by a matrix moves it. Every 2 × 2 matrix stretches, squashes, rotates, shears or flips the whole plane, and keeps straight lines straight. That is why it is called a linear transformation .
- **Identity, determinant and inverse** The identity matrix I has 1s on its diagonal and 0s elsewhere. It does nothing: AI = IA = A, just like multiplying by 1. The inverse A^{-1} undoes A: $A^{-1}A = I$.
- **Solving many equations at once** A system of linear equations is one matrix equation Ax = b . The matrix A holds the numbers in front of the unknowns, x is the vector of unknowns, and b is the right hand side.
- **Rank and special matrices** The rank of a matrix is the number of truly independent columns (or rows, it is always the same number). It says how many dimensions the matrix really uses. A 1000 × 50 data matrix with rank 30 has 20 columns that are combinations of others.

FORMULAS

Addition

Exam favourite

$$(A + B)_{ij} = a_{ij} + b_{ij}$$

Only for equal shapes.

Lesson: Adding, scaling and transposing

Transpose

$$(A^T)_{ij} = a_{ji}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Lesson: Adding, scaling and transposing

Element wise (Hadamard) product

$$(A \odot B)_{ij} = a_{ij} \times b_{ij}$$

Multiply matching entries. This is NOT matrix multiplication.

Lesson: Adding, scaling and transposing

Matrix product

Exam favourite

$$(AB)_{ij} = \Sigma_{k}\, a_{ik}\, b_{kj}$$

Row i of A dotted with column j of B.

Lesson: Matrix multiplication, step by step

Shape rule

$$(m \times \mathbf{n}) \cdot (\mathbf{n} \times p) = m \times p$$

The inner sizes must match. The outer sizes give the answer shape.

Lesson: Matrix multiplication, step by step

2 × 2 determinant

Exam favourite

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

How much A scales area. A negative sign means it also flips the plane over.

Lesson: Identity, determinant and inverse

2 × 2 inverse

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Swap a and d, change the signs of b and c, divide by the determinant.

Lesson: Identity, determinant and inverse

Linear regression in matrix form (normal equations)

Exam favourite

$$\beta = (X^T X)^{-1} X^T y$$

X holds the inputs (with a column of 1s for the intercept), y the targets, β the best weights. Practice problem 4 builds this by hand.

Lesson: Solving many equations at once

Rank and the lost directions

Exam favourite

rank + (number of lost directions) = number of columns

A 3 × 3 matrix of rank 2 squashes one whole direction to zero. That lost direction is called the null space.

Lesson: Rank and special matrices

TABLES

Table: object · shape · example

Object	Shape	Example
Scalar	a single number	learning rate 0.01
Vector	n numbers	one house [3, 120, 15]
Matrix	m × n	all houses, or the weights of one layer
Tensor	three or more directions	a batch of colour images: 32 × 3 × 224 × 224

Lesson: What a matrix is

Table: question · maths · numpy or pytorch

Question	Maths	NumPy or PyTorch
How big is it?	m × n	A.shape gives (m, n)
One entry	a_{ij}	A[i, j], counting from 0
A whole row	row i	A[i, :] or A[i]
A whole column	column j	A[:, j]
Number of examples	m (rows of X)	X.shape[0], also len(X)

Lesson: What a matrix is

Table: property · true or false · why it matters

Property	True or false	Why it matters
AB = BA	**false** in general	order matters. The program above shows AB and BA are different
(AB)C = A(BC)	true (associative)	you can group a chain of layers in any way you like
A(B + C) = AB + AC	true (distributive)	used all the time in derivations
cost of n × n times n × n	about n^3 multiplications	why big models need GPUs

Lesson: Matrix multiplication, step by step

Table: view · what ab is · where it helps

View	What AB is	Where it helps
dot products	entry (i, j) is row i of A dotted with column j of B	computing by hand, attention scores QK^T
columns	column j of AB is A times column j of B	a layer applied to many inputs at once
rows	row i of AB is row i of A times B	each example (a row of X) passes through the same weights
outer products	AB is the sum over k of (column k of A) times (row k of B)	low rank ideas such as LoRA and the SVD

Lesson: Matrix multiplication, step by step

Table: matrix · what it does

Matrix	What it does
$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$	scales everything by 2
$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$	rotates by 90° anticlockwise
$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$	shears sideways, like pushing the top of a stack of cards
$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	reflects in the x axis
$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	rotates by any angle θ

Lesson: A matrix is a transformation

Table: what the last rows look like · what it means · how many answers

What the last rows look like	What it means	How many answers
a staircase with no zero on the diagonal	every equation adds new information	exactly one
a row of all zeros = 0	one equation repeated another	infinitely many
a row of zeros = a non zero number	the equations contradict each other (0 = 5)	none

Lesson: Solving many equations at once

Table: special matrix · what it looks like · why it is handy

Special matrix	What it looks like	Why it is handy
Diagonal	non zero entries only on the diagonal	multiplying just scales each coordinate
Symmetric	$A = A^T$	covariance matrices are symmetric, and their eigenvalues are real (Chapter 5)
Orthogonal	$Q^TQ = I$	rotations and reflections, which keep lengths and angles
Triangular	zeros above or below the diagonal	easy to solve by back substitution
Sparse	mostly zeros	stored and multiplied cheaply, as in recommendation data

Lesson: Rank and special matrices

TIPS AND TRAPS

REMEMBER

Say the shape out loud before you write any matrix code: "X is examples by features, W is features by outputs, so XW is examples by outputs". Most shape bugs disappear with this habit.

Lesson: What a matrix is

REMEMBER Transpose twice and you are back where you started: $(A^T)^T = A$. And for products the order reverses: $(AB)^T = B^T A^T$, like taking off socks and shoes in the reverse order.

Lesson: Adding, scaling and transposing

IN THE EXAM In NumPy, `A * B` is the element wise product and `A @ B` is matrix multiplication. Mixing them up gives wrong answers without any error message when the shapes happen to fit.

Lesson: Adding, scaling and transposing

REMEMBER "Row times column, row times column". Put your left index finger on a row of A and your right index finger on a column of B. Move both fingers together, multiplying as you go.

Lesson: Matrix multiplication, step by step

REMEMBER A negative determinant means the parallelogram is flipped over, like looking at the square in a mirror. The size of the area is still $|\det|$.

Lesson: Identity, determinant and inverse

COMMON MISTAKE If $\det A = 0$ the matrix squashes the plane flat onto a line or a point. Information is lost, so nothing can undo it. Such a matrix is called **singular** and has no inverse.

Lesson: Identity, determinant and inverse

REMEMBER Rank is never more than the smaller of the two sizes. A 3×7 matrix has rank at most 3.

Lesson: Rank and special matrices

2.3 Eigenvalues, SVD and PCA

Some directions are special for a matrix: it only stretches them. Hunt for these eigenvectors with an animation, find them by hand, watch the SVD turn a circle into an ellipse, and run PCA step by step to squash data into fewer numbers.

KEY IDEAS

- **Eigenvectors: the directions a matrix keeps** Chapter 4 showed that a matrix moves every arrow in the plane. Most arrows get turned to a new direction. A few special arrows only get longer, shorter or flipped , and stay on their own line.
- **Finding eigenvalues by hand**
- **Rules that save time** Eigenvalues come with a handful of rules that let you check your work in seconds and that explain why they are so useful.
- **Power iteration and PageRank** Real matrices have thousands or millions of rows, so nobody solves a characteristic equation. A simple and famous method is power iteration : start with any vector, multiply by A, scale it back to length 1, and repeat.
- **Singular value decomposition (SVD)** Eigenvectors need a square matrix, and even then they might not exist. The SVD works for every matrix of any shape. It says each matrix is three simple moves in a row: a rotation, a stretch along the axes, and another rotation.
- **Principal component analysis (PCA)** PCA finds the directions in which your data spreads out the most. Keep the top few and you can describe 100 features with, say, 10 numbers, while keeping most of the information. It is eigenvectors applied to data.
- **Where machine learning uses all this** Eigenvalues and the SVD appear wherever a model needs the "main directions" of something. Here is a quick map, which later chapters revisit.

FORMULAS

The eigen equation

Exam favourite

$$A\,v = \lambda\,v \quad \text{with } v \neq 0$$

v is the eigenvector (the direction that survives). λ is the eigenvalue (how much it is stretched). λ = 2 doubles v, λ = 0.5 halves it, λ = −1 flips it.

Lesson: Eigenvectors: the directions a matrix keeps

Characteristic equation

Exam favourite

$$\det(A - \lambda I) = 0$$

Solve for λ. A 2 × 2 matrix gives a quadratic, a 3 × 3 gives a cubic.

Lesson: Finding eigenvalues by hand

Shortcut for 2 × 2

$$\lambda^2 - (\text{trace})\,\lambda + \det = 0$$

trace = a + d, det = ad − bc. Then $\lambda = \dfrac{\text{trace} \pm \sqrt{(\text{trace}^2 - 4\,\det)}}{2}$

Lesson: Finding eigenvalues by hand

Diagonalisation (eigendecomposition)

Exam favourite

$$A = P\,D\,P^{-1} \quad \text{and} \quad A^n = P\,D^n\,P^{-1}$$

P has the eigenvectors as its columns. D is diagonal with the eigenvalues. For a symmetric matrix P is orthogonal, so $P^{-1} = P^T$.

Lesson: Rules that save time

Power iteration

Exam favourite

$$v \leftarrow A\,v \,/\, \|A\,v\| \qquad \lambda \approx v^T A\,v$$

The second formula (the Rayleigh quotient) reads off the eigenvalue once v has settled.

Lesson: Power iteration and PageRank

Speed of power iteration

$$\text{error after k steps} \approx |\lambda_2 \div \lambda_1|^k$$

With $\lambda_1 = 5$ and $\lambda_2 = 2$ the error shrinks by 0.4 each step. With $\lambda_1 = 1$ and $\lambda_2 = 0.99$ it shrinks by only 1% each step, which is very slow.

Lesson: Power iteration and PageRank

Singular value decomposition

Exam favourite

$$A = U\,\Sigma\,V^T$$

For an m × n matrix: U is m × m and orthogonal, Σ is m × n with the singular values $\sigma_1 \geq \sigma_2 \geq \ldots \geq 0$ on its diagonal, and V is n × n and orthogonal.

Lesson: Singular value decomposition (SVD)

Where the parts come from

$$\sigma_i = \sqrt{(\text{eigenvalues of } A^TA)}$$

The columns of V are the eigenvectors of A^TA. Then $u_i = A\,v_i / \sigma_i$.

Lesson: Singular value decomposition (SVD)

Best rank k copy (Eckart and Young)

$$A_k = \sigma_1 u_1 v_1^T + \ldots + \sigma_k u_k v_k^T$$

Keeping the k biggest singular values gives the closest possible rank k matrix. The error is set by the singular values you dropped.

Lesson: Singular value decomposition (SVD)

Storage for a rank k copy

$$k \times (m + n + 1) \text{ numbers}$$

k columns of U (m each), k rows of V^T (n each) and k singular values, instead of m × n.

Lesson: Singular value decomposition (SVD)

Share of the "energy" kept

$$\frac{\sigma_1^2 + \ldots + \sigma_k^2}{\sigma_1^2 + \ldots + \sigma_r^2}$$

The squared singular values add up to the sum of all squared entries of the matrix. This share says how much of it the copy keeps.

Lesson: Singular value decomposition (SVD)

Covariance of two features

Exam favourite

$$\text{cov}(x, y) = \frac{\Sigma\,(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Positive: they rise together. Negative: one rises as the other falls. The diagonal of C holds the variances.

Lesson: Principal component analysis (PCA)

Explained variance ratio

$$\text{share of PC k} = \frac{\lambda_k}{\lambda_1 + \lambda_2 + \ldots + \lambda_n}$$

Keep enough components to reach about 90 to 95 percent.

Lesson: Principal component analysis (PCA)

Condition number

Exam favourite

$$\kappa(A) = \sigma\max / \sigma\min$$

Large κ means a tiny change in the input can cause a large change in the answer. Near duplicate features make κ huge, which is why ridge regression adds λI to X^TX.

Lesson: Where machine learning uses all this

TABLES

Table: rule · in symbols · example with [[2, 1], [1, 2]]

Rule	In symbols	Example with [[2, 1], [1, 2]]
Sum of eigenvalues = trace	$\lambda_1 + \lambda_2 + \ldots = a_{11} + a_{22} + \ldots$	3 + 1 = 4 = 2 + 2
Product of eigenvalues = determinant	$\lambda_1 \times \lambda_2 \times \ldots = \det A$	3 × 1 = 3 = det A
Powers	$A^n v = \lambda^n v$	A^{10} [1, 1] = 3^{10} [1, 1] = 59049 [1, 1]
Inverse	$A^{-1} v = (1/\lambda) v$	eigenvalues of A^{-1} are 1/3 and 1
Symmetric matrix	$A = A^T$ gives real λ and perpendicular v	[1, 1] · [1, −1] = 0

Lesson: Rules that save time

Table: use of svd · what it does

Use of SVD	What it does
Image compression	keep the top k singular values of the picture matrix
Recommendations	the Netflix Prize matrix of users × films was filled in with low rank factorisations like the SVD
Search (latent semantic analysis)	SVD of a words × documents table groups words with related meanings
Pseudo inverse and least squares	$A^+ = V \Sigma^+ U^T$ solves regression even when X^TX has no inverse

Lesson: Singular value decomposition (SVD)

Table: step · what you do · why

Step	What you do	Why
1. Centre	subtract the mean of every feature	so the cloud sits around the origin
2. Covariance	$C = X^TX / (n - 1)$ for the centred X	C says how features vary together
3. Eigen	find the eigenvalues and eigenvectors of C	eigenvectors are the principal directions
4. Sort and keep	order by eigenvalue, keep the top k	eigenvalue = variance along that direction
5. Project	scores = X × (top k eigenvectors)	each row becomes k numbers

Lesson: Principal component analysis (PCA)

Table: area · what the eigen idea gives

Area	What the eigen idea gives
Dimensionality reduction	PCA compresses features, speeds up training and lets you plot high dimensional data in 2D
Optimisation (Chapter 8)	the eigenvalues of the Hessian describe the shape of the loss surface. The ratio λmax / λmin, the condition number, says how slow gradient descent will be
Spectral clustering	eigenvectors of a graph Laplacian split points into groups that ordinary k means misses
Recommenders	a low rank factorisation of the users × items table predicts missing ratings
Large language models	LoRA fine tuning adds a low rank update, and SVD is used to compress weight matrices
Stability	eigenvalues of recurrent weight matrices above 1 make signals explode, below 1 make them vanish

Lesson: Where machine learning uses all this

Table: eigenvalue or singular value pattern · what it tells you

Eigenvalue or singular value pattern	What it tells you
a few large values, then a sharp drop	the data really lives in a low dimensional space, PCA will work well
values falling slowly and evenly	information is spread out, compression will lose a lot
one value close to 0	two or more features are nearly dependent, watch out for unstable weights

Lesson: Where machine learning uses all this

TIPS AND TRAPS

REMEMBER "Eigen" is German for "own". An eigenvector is a direction the matrix owns, the one it does not turn.
Lesson: Eigenvectors: the directions a matrix keeps

REMEMBER Any multiple of an eigenvector is also an eigenvector. [1, 1], [2, 2] and [−5, −5] all point along the same line. Libraries usually scale it to length 1, so NumPy gives [0.7071, 0.7071] for [1, 1].
Lesson: Finding eigenvalues by hand

REMEMBER Triangular or diagonal matrix? The eigenvalues are simply the numbers on the diagonal. No equation needed.
Lesson: Finding eigenvalues by hand

REMEMBER The two rows of $A - \lambda I$ are always multiples of each other (that is what det = 0 means). So you only ever need to read one row to find v.
Lesson: Finding eigenvalues by hand

REMEMBER Two quick checks after finding eigenvalues: do they add up to the trace, and multiply to the determinant? If not, there is an arithmetic slip.
Lesson: Rules that save time

REMEMBER Remember "rotate, stretch, rotate". V^T turns the input, Σ stretches along the axes, U turns the result. The circle of all unit inputs always becomes an ellipse whose half axes are the singular values.
Lesson: Singular value decomposition (SVD)

COMMON MISTAKE Scale your features first. If height is in millimetres and weight in tonnes, the millimetre column has a huge variance only because of its unit, and PCA will think it is the most important direction. Standardise each column (subtract the mean, divide by the standard deviation) before PCA.
Lesson: Principal component analysis (PCA)

REMEMBER Big eigenvalue = important direction. Tiny eigenvalue = direction you can usually drop. Zero eigenvalue = a direction the matrix throws away completely.

Lesson: Where machine learning uses all this

STEP 3 Calculus

3.1 Derivatives and Integrals

A derivative is a speedometer for any function. Watch a secant line become a tangent, learn the rules that make derivatives quick, master the chain rule behind backpropagation, find peaks and valleys, and fill areas with rectangles to meet the integral.

KEY IDEAS

- **Rate of change and slope** A derivative answers one question: how fast is something changing right now? A car that travels 120 km in 2 hours has an average speed of 60 km/h, but the speedometer shows the speed at this exact moment. The derivative is the speedometer.
- **Limits and the derivative from first principles** A limit says what value a formula approaches as the input gets close to some number, even if you cannot plug that number in. You cannot set h = 0 in (f(a + h) − f(a)) ÷ h because that divides by zero, but you can ask what happens as h shrinks.
- **The rules of differentiation** Nobody works from first principles every time. A small set of rules covers almost every function you will meet in machine learning.
- **The chain rule** Most functions in machine learning are functions inside functions. A neuron computes z = wx + b, then a = σ(z), then a loss L(a). The chain rule says how a small change at the start ripples through every link.
- **Peaks, valleys and the second derivative** Where a smooth curve reaches a peak or the bottom of a valley, it is flat for a moment, so its slope is 0. Setting f'(x) = 0 finds these critical points . The second derivative f"(x), the slope of the slope, tells which kind it is.
- **Integrals: adding up tiny pieces** An integral is the opposite job of a derivative. A derivative turns distance into speed. An integral turns speed back into distance, by adding up "speed × tiny time" over and over. In pictures, it is the area under a curve .
- **Why derivatives make machines learn** Training a model means making a loss number small. The derivative of the loss with respect to a weight says which way to nudge that weight and how strongly. If ∂L/∂w is positive, increasing w raises the loss, so decrease it. If it is negative, increase w.

FORMULAS

Average rate of change (slope of a secant)

Exam favourite

$$\frac{f(b) - f(a)}{b - a}$$

Rise over run between two points on the curve.

Lesson: Rate of change and slope

Instant rate of change (slope of the tangent)

$$f'(a) = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

Slide the second point closer and closer to the first. The slope settles on one number.

Lesson: Rate of change and slope

Power rule

Exam favourite

$$d/dx\ x^n = n\ x^{n-1}$$

Bring the power down, then lower it by one. x^5 becomes $5x^4$.

Lesson: The rules of differentiation

Constant and sum rules

$$(c \cdot f)' = c \cdot f' \quad (f + g)' = f' + g'$$

Constants ride along. Differentiate each term on its own.

Lesson: The rules of differentiation

Product rule

$$(f \cdot g)' = f'g + fg'$$

First times slope of second, plus second times slope of first.

Lesson: The rules of differentiation

Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

"Low d high minus high d low, over low squared".

Lesson: The rules of differentiation

Chain rule

Exam favourite

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

If y depends on u and u depends on x, multiply the slopes along the chain. Outside slope times inside slope.

Lesson: The chain rule

Convex function

Exam favourite

$$f''(x) \geq 0 \text{ everywhere}$$

The curve is a single bowl, so any valley you find is THE lowest point. Mean squared error for linear regression is convex, which is why it always trains to the same answer.

Lesson: Peaks, valleys and the second derivative

Definite integral as a sum of strips

Exam favourite

$$\int_a^b f(x)\,dx \approx \Sigma\, f(x_i)\,\Delta x$$

Cut [a, b] into thin strips of width Δx and add up height × width. More strips, better answer.

Lesson: Integrals: adding up tiny pieces

Fundamental theorem of calculus

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

where F' = f. Find a function whose slope is f, then subtract its values at the ends.

Lesson: Integrals: adding up tiny pieces

Power rule backwards

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$$

Raise the power by one, divide by the new power (n ≠ −1).

Lesson: Integrals: adding up tiny pieces

Average value of f on [a, b]

$$\frac{1}{b-a}\int_a^b f(x)\,dx$$

The average of x² from 0 to 2 is (8 ÷ 3) ÷ 2 = 4 ÷ 3 ≈ 1.333.

Lesson: Integrals: adding up tiny pieces

TABLES

Table: notation · read it as · who uses it

Notation	Read it as	Who uses it
f'(x)	f prime of x	Lagrange, most school books
dy/dx	dee y by dee x	Leibniz, physics and machine learning papers
∂L/∂w	partial L by partial w	when there are many inputs (Chapter 7)

Lesson: Rate of change and slope

Table: function f(x) · derivative f'(x) · note

Function f(x)	Derivative f'(x)	Note
c (a constant)	0	a flat line has no slope
x^n	$n\,x^{n-1}$	works for any power, also √x = x^(1/2)
e^x	e^x	the only function that is its own slope
a^x	$a^x \ln a$	for example 2^x has slope $2^x \times 0.693$
ln x	1/x	used in every log loss
sin x	cos x	
cos x	−sin x	

Lesson: The rules of differentiation

Table: sign · what the curve does · picture

Sign	What the curve does	Picture
f'(x) > 0	rising	↗
f'(x) < 0	falling	↘
f'(x) = 0 and f''(x) > 0	local minimum, bends upward like a cup	∪
f'(x) = 0 and f''(x) < 0	local maximum, bends downward like a cap	∩
f''(x) changes sign	inflection point, the bend flips	∫

Lesson: Peaks, valleys and the second derivative

Table: activation · formula · derivative · comment

Activation	Formula	Derivative	Comment
Sigmoid	$\sigma(z) = 1 \div (1 + e^{-z})$	$\sigma(1 - \sigma)$	at most 0.25, can vanish
Tanh	$\tanh z$	$1 - \tanh^2 z$	at most 1
ReLU	$\max(0, z)$	1 if $z > 0$, else 0	cheap, does not vanish for positive z
Softplus	$\ln(1 + e^z)$	$\sigma(z)$	a smooth ReLU

Lesson: Why derivatives make machines learn

TIPS AND TRAPS

REMEMBER The **central difference** $(f(x + h) - f(x - h)) \div 2h$ looks both ways and is far more accurate than the one sided version. Machine learning engineers use it for **gradient checking**: comparing their hand written derivative with this numerical one to catch bugs.

Lesson: Limits and the derivative from first principles

REMEMBER A derivative is just "rise over run" for a run so short you cannot see it.

Lesson: Limits and the derivative from first principles

COMMON MISTAKE Do not make h too tiny on a computer. With $h = 10^{-15}$ the subtraction $f(x + h) - f(x)$ loses almost all its digits to rounding. Something like $h = 10^{-5}$ is a safe middle ground.

Lesson: Limits and the derivative from first principles

REMEMBER Product rule chant: "left d right plus right d left". Quotient rule chant: "low d high minus high d low, over the square of what is below".

Lesson: The rules of differentiation

REMEMBER Peel an onion: differentiate the outside layer, keep the inside untouched, then multiply by the slope of the inside.

Lesson: The chain rule

REMEMBER Slope positive, step left. Slope negative, step right. Slope zero, you are at the bottom (or a flat spot).

Lesson: Why derivatives make machines learn

3.2 Gradients and Backpropagation

A model has many weights, so its loss is a surface in many directions. Slice it with partial derivatives, follow gradient arrows downhill on a contour map, meet the Jacobian and Hessian, and run backpropagation through a graph one node at a time.

KEY IDEAS

- **Partial derivatives: one knob at a time** Real things depend on many inputs. A house price depends on its size and its age. A model loss depends on millions of weights. A partial derivative asks how the output changes when you turn one input and hold every other input still.
- **The gradient: the steepest way uphill** Collect all the partial derivatives into one vector and you get the gradient , written ∇f (said "grad f" or "del f"). It is the most important object in machine learning.
- **Directional derivatives and linear approximation** The gradient answers "how fast does f change if I step in any direction u?" Just take a dot product. And near a point, every smooth surface looks almost flat, so a tangent plane predicts nearby values.
- **Jacobian and Hessian matrices** Two matrices organise all the derivatives of many inputs at once.
- **The chain rule on a computational graph** Any formula can be drawn as a computational graph : input boxes flow through small operations (add, multiply, sigmoid) to the output. The forward pass computes values left to right.
- **Backpropagation through a neuron** Now put it all together for one neuron. Inputs x = [1, 2], weights w = [0.5, −0.3], bias b = 0.1, a sigmoid, and the squared error loss against the target y = 1.
- **Gradients for whole datasets** A real loss is the average over many examples, so its gradient is the average of the per example gradients. With matrices, the whole batch is done in one line.

FORMULAS

Partial derivative

Exam favourite

$$\frac{\partial f}{\partial x} = \text{slope of f when only x moves}$$

The curly ∂ means "partial". Treat every other letter as a fixed number, then use the ordinary rules from Chapter 6.

Lesson: Partial derivatives: one knob at a time

Gradient

Exam favourite

$$\nabla f = \begin{pmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \\ \vdots \\ \partial f/\partial x_n \end{pmatrix}$$

It points in the direction of steepest increase. Its length says how steep that climb is. Minus the gradient is the steepest way down.

Lesson: The gradient: the steepest way uphill

Directional derivative

Exam favourite

$$D_u f = \nabla f \cdot u$$

u is a unit vector. It is largest when u points along ∇f, equal to $\|\nabla f\|$, and zero along a contour.

Lesson: Directional derivatives and linear approximation

Linear (first order) approximation

$$f(x + \Delta) \approx f(x) + \nabla f(x) \cdot \Delta$$

The tangent plane. Good for small steps, which is why learning rates are kept small.

Lesson: Directional derivatives and linear approximation

Jacobian (first derivatives of many outputs)

Exam favourite

$$J = \begin{pmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{pmatrix}$$

Row i holds the gradient of output i. For a layer z = Wx + b the Jacobian ∂z/∂x is just W.

Lesson: Jacobian and Hessian matrices

Hessian (second derivatives of one output)

$$H = \begin{pmatrix} \partial^2 f/\partial x^2 & \partial^2 f/\partial x \partial y \\ \partial^2 f/\partial y \partial x & \partial^2 f/\partial y^2 \end{pmatrix}$$

It describes the curvature of the surface. It is symmetric, because the order of differentiating does not matter for smooth f.

Lesson: Jacobian and Hessian matrices

Multivariable chain rule

Exam favourite

$$\frac{\partial L}{\partial x} = \Sigma_{\text{paths}} \text{ (product of local slopes along the path)}$$

If x affects L through several routes, add up the contributions of every route.

Lesson: The chain rule on a computational graph

Mean squared error for linear regression

Exam favourite

$$L(w) = \frac{1}{n} \|Xw - y\|^2$$

X holds one row per example, w the weights, y the targets.

Lesson: Gradients for whole datasets

Its gradient

$$\nabla L = \frac{2}{n} X^T(Xw - y)$$

Errors weighted by each feature, then averaged. Setting it to zero gives the normal equations of Chapter 4.

Lesson: Gradients for whole datasets

TABLES

Table: function · ∂/∂x · ∂/∂y

Function	∂/∂x	∂/∂y
3x + 5y	3	5
xy	y	x
$x^2 + y^2$	2x	2y
e^(xy)	y e^(xy)	x e^(xy)
$w_1x_1 + w_2x_2 + b$ (in w_1)	x_1 (the input that w_1 multiplies)	not used

Lesson: Partial derivatives: one knob at a time

Table: fact about ∇f · why it matters

Fact about ∇f	Why it matters
points straight uphill	so −∇f points straight downhill, the way to reduce a loss
is perpendicular to the contour lines	contours are paths of equal height, and uphill is at right angles to them
has length 0 at peaks, valleys and saddles	a zero gradient means training has stopped moving
has one entry per input	a model with 7 billion weights has a gradient with 7 billion entries

Lesson: The gradient: the steepest way uphill

Table: hessian at a flat point · shape · test for 2 inputs

Hessian at a flat point	Shape	Test for 2 inputs
all eigenvalues positive	bowl: a minimum	$\det H > 0$ and $\partial^2 f/\partial x^2 > 0$
all eigenvalues negative	cap: a maximum	$\det H > 0$ and $\partial^2 f/\partial x^2 < 0$
mixed signs	saddle: up one way, down another	$\det H < 0$

Lesson: Jacobian and Hessian matrices

Table: node · forward · backward rule

Node	Forward	Backward rule
add: $q = a + b$	adds	passes the incoming gradient unchanged to both inputs
multiply: $q = a \times b$	multiplies	sends gradient $\times$ b to a, and gradient $\times$ a to b (swap)
max: $q = \max(a, b)$	picks the bigger	sends the whole gradient to the winner only, like ReLU
sigmoid: $q = \sigma(a)$	squashes	multiplies the gradient by $q(1 - q)$

Lesson: The chain rule on a computational graph

Table: term · meaning

Term	Meaning
Batch gradient	average gradient over the whole dataset, accurate but slow
Mini batch gradient	average over 32 to 1024 random examples, the standard choice
Stochastic gradient	gradient from one example, very noisy but cheap
Autograd	the library builds the computational graph and runs backprop for you

Lesson: Gradients for whole datasets

TIPS AND TRAPS

REMEMBER Picture standing on a hillside in fog. You cannot see the valley, but you can feel which way the ground tilts under your feet. That tilt is the gradient, and gradient descent is taking a small step the other way.

Lesson: The gradient: the steepest way uphill

REMEMBER Add nodes copy, multiply nodes swap, max nodes route, and every node multiplies by its local slope.

Lesson: The chain rule on a computational graph

REMEMBER Shapes must match: the gradient of a loss with respect to a weight matrix always has the same shape as that weight matrix.

Lesson: Gradients for whole datasets

3.3 Gradient Descent and Optimisation

Training a model is walking downhill on its loss. Run gradient descent step by step, find the learning rate that is just right, see why mini batches work, race plain gradient descent against momentum and Adam, and solve problems with constraints using Lagrange multipliers.

KEY IDEAS

- **Gradient descent: walk downhill** Training a model means finding the weights that make the loss as small as possible. Gradient descent does it the way a skier finds the bottom of a valley in fog: feel the slope under your feet, take a small step downhill, and repeat.
- **The learning rate: not too small, not too big** The learning rate is the most important setting in training. Too small and training crawls. Too big and every step overshoots the valley floor, bouncing from side to side or flying off completely.
- **Stochastic and mini batch gradient descent** The true gradient averages over every training example. With a million examples, that is a million calculations for one step. Stochastic gradient descent (SGD) estimates the gradient from a small random sample instead, and takes many more, cheaper steps.
- **Momentum, RMSProp and Adam** Plain gradient descent struggles in long narrow valleys. The steep walls make it zig zag while the gentle floor makes it crawl.

- **Convexity, local minima and saddles** For a convex loss there is one valley, so gradient descent with a sensible learning rate always reaches the best answer. Linear regression, logistic regression and SVMs have convex losses.
- **Optimising with constraints: Lagrange multipliers** Sometimes the answer must obey a rule: a fixed budget, a fixed length of fence, probabilities that add to 1.
- **Training tricks: schedules, clipping and stopping** Real training runs add a few simple tools around the optimiser.

FORMULAS

The gradient descent update

Exam favourite

$$w \leftarrow w - \eta \nabla L(w)$$

w are the weights, ∇L is the gradient of the loss (Chapter 7) and η (eta) is the learning rate, the size of each step.

Lesson: Gradient descent: walk downhill

Progress per step on a smooth loss

$$L(w - \eta \nabla L) \approx L(w) - \eta \, \|\nabla L\|^2$$

From the linear approximation of Chapter 7. As long as η is small enough, every step lowers the loss by roughly η times the squared gradient length.

Lesson: Gradient descent: walk downhill

Stability rule for a bowl L = ½ a w²

Exam favourite

$$w \leftarrow (1 - \eta a)\, w \quad \text{converges when } 0 < \eta < \frac{2}{a}$$

a is the curvature (second derivative). Steeper bowls need smaller steps. For many weights, a is the largest Hessian eigenvalue.

Lesson: The learning rate: not too small, not too big

Mini batch gradient

Exam favourite

$$g = \frac{1}{B} \sum_{i \in \text{batch}} \nabla L_i(w)$$

An unbiased estimate of the full gradient. Its noise shrinks like 1/√B.

Lesson: Stochastic and mini batch gradient descent

Epoch

$$\text{updates per epoch} = \frac{n}{B}$$

One epoch is one full pass over the data. 50000 images with B = 100 gives 500 updates per epoch.

Lesson: Stochastic and mini batch gradient descent

Noise in a mini batch gradient

$$\text{spread} \propto \frac{\sigma}{\sqrt{B}}$$

σ is how much the per example gradients disagree. Four times the batch size halves the noise, the same √n rule as the standard error in Chapter 10.

Lesson: Stochastic and mini batch gradient descent

Momentum

Exam favourite

$$v \leftarrow \beta v + \nabla L \quad w \leftarrow w - \eta v$$

v is a running sum of gradients, like a heavy ball gathering speed. Zig zags cancel out, steady directions add up. β ≈ 0.9.

Lesson: Momentum, RMSProp and Adam

RMSProp

$$s \leftarrow \rho s + (1 - \rho)\, g^2 \quad w \leftarrow w - \eta \, g \div \sqrt{s}$$

Divide by the typical size of recent gradients, so steep directions get smaller steps and flat ones bigger.

Lesson: Momentum, RMSProp and Adam

Adam (momentum plus RMSProp)

$$m \leftarrow \beta_1 m + (1 - \beta_1) g \quad v \leftarrow \beta_2 v + (1 - \beta_2) g^2$$

$$w \leftarrow w - \eta \, \hat{m} \div (\sqrt{\hat{v}} + \varepsilon)$$

$\hat{m} = m/(1 - \beta_1^t)$ and $\hat{v} = v/(1 - \beta_2^t)$ fix the start up bias. Defaults: $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$.

Lesson: Momentum, RMSProp and Adam

Condition number of the loss

Exam favourite

$$\kappa = \lambda\text{max} \div \lambda\text{min of the Hessian}$$

Large κ means a long narrow valley. Plain gradient descent needs roughly κ steps to make progress, which is why feature scaling and adaptive optimisers matter.

Lesson: Convexity, local minima and saddles

Lagrange condition

Exam favourite

$$\nabla f = \lambda \nabla g \quad \text{together with} \quad g(x) = c$$

f is what you optimise, g(x) = c is the rule. The multiplier λ tells how much the best value of f improves per extra unit of c.

Lesson: Optimising with constraints: Lagrange multipliers

Cosine decay

Exam favourite

$$\eta(t) = \eta_{max} \times ½ \, (1 + \cos(\pi \, t \div T))$$

Starts at the maximum and glides smoothly to 0 at step T.

Lesson: Training tricks: schedules, clipping and stopping

Gradient clipping by norm

$$g \leftarrow g \times \min(1, \frac{c}{\|g\|})$$

Keeps the direction, caps the length at c.

Lesson: Training tricks: schedules, clipping and stopping

Warm up then cosine

$$\eta(t) = \eta_{max} \times \frac{t}{T_w} \text{ for } t < T_w, \text{ then } \eta_{max} \times ½(1 + \cos(\pi \frac{t - T_w}{T - T_w}))$$

T_w is the warm up length and T the total number of steps.

Lesson: Training tricks: schedules, clipping and stopping

TABLES

Table: stopping rule · check · typical setting

Stopping rule	Check	Typical setting
fixed budget	stop after a set number of steps or epochs	10 epochs, or 100000 steps
small gradient	stop when $\|\nabla L\|$ is tiny, since that means the ground is flat	$\|\nabla L\| < 10^{-6}$
small improvement	stop when the loss barely changes between checks	less than 0.01% better
early stopping	stop when the validation loss has not improved for a while	patience of 5 epochs

Lesson: Gradient descent: walk downhill

Table: learning rate · what happens · loss curve

Learning rate	What happens	Loss curve
far too small	tiny steps, training takes forever	falls very slowly
about right	big steps at first, smaller near the bottom	falls fast, then flattens
a bit too big	overshoots, zig zags across the valley	falls with wobbles
far too big	every step lands higher than the last	shoots up, often to NaN

Lesson: The learning rate: not too small, not too big

Table: shape of the loss curve · diagnosis · action

Shape of the loss curve	Diagnosis	Action
almost flat from the start	learning rate far too small	multiply η by 10
smooth fast drop, then flat	about right	keep it
drops, but saw tooth wobbles	a little too big	halve η, or add momentum
rises, or jumps to NaN	far too big, diverging	divide η by 10, add gradient clipping

Lesson: The learning rate: not too small, not too big

Table: version · examples per step · pros · cons

Version	Examples per step	Pros	Cons
Batch GD	all n	exact gradient, smooth path	slow, needs all data in memory
Stochastic GD	1	very cheap steps, noise can escape bad spots	very noisy path
Mini batch GD	typically 32 to 1024	fast on GPUs, mild noise	one more setting to choose

Lesson: Stochastic and mini batch gradient descent

Table: optimiser · typical use

Optimiser	Typical use
SGD with momentum	image models such as ResNets, often with the best final accuracy
Adam	the default for most projects, quick to get working
AdamW	Adam with weight decay done properly, standard for transformers and large language models

Lesson: Momentum, RMSProp and Adam

Table: feature · what it is · how training copes

Feature	What it is	How training copes
Global minimum	the lowest point of all	the ideal, rarely guaranteed for neural networks
Local minimum	a valley that is not the lowest	in big networks most are nearly as good as the global one
Saddle point	flat, but downhill in some directions	noise from SGD and momentum push the weights off
Plateau	a wide nearly flat region	Adam helps by enlarging tiny gradients
Cliff	a sudden very steep wall	gradient clipping caps the step size

Lesson: Convexity, local minima and saddles

Table: case · what happens · multiplier

Case	What happens	Multiplier
the limit is not reached	the best point is inside the allowed region, so $\nabla f = 0$ there as usual	$\lambda = 0$, the rule costs nothing
the limit is reached	the best point sits on the boundary, and the Lagrange rule $\nabla f = \lambda \nabla g$ holds	$\lambda > 0$, loosening the rule would help

Lesson: Optimising with constraints: Lagrange multipliers

Table: tool · what it does · why

Tool	What it does	Why
Learning rate warm up	start with a tiny learning rate and grow it over the first steps	early gradients are wild while the Adam averages are still empty
Step or cosine decay	lower the learning rate as training goes on	big steps to explore, small steps to settle into the valley
Gradient clipping	if $\|g\| > c$, scale g down to length c	stops one huge gradient from wrecking the weights
Early stopping	stop when the validation loss stops improving	prevents overfitting and saves time
Weight decay	shrink every weight a little each step: $w \leftarrow w - \eta\lambda w$	the L2 regularisation of Chapter 12

Lesson: Training tricks: schedules, clipping and stopping

TIPS AND TRAPS

REMEMBER Gradient descent never needs to see the whole landscape. Like a ball rolling downhill, it only needs the slope where it is standing right now.

Lesson: Gradient descent: walk downhill

REMEMBER A common recipe: try learning rates spaced by factors of 10 (0.1, 0.01, 0.001), watch the loss for a few hundred steps, and keep the largest one that still falls smoothly. Adam usually starts at 0.001.

Lesson: The learning rate: not too small, not too big

REMEMBER The first Adam step always moves each weight by about η, whatever the size of the gradient, because $\hat{m} \div \sqrt{\hat{v}} = g \div |g| = \pm 1$ at $t = 1$.

Lesson: Momentum, RMSProp and Adam

REMEMBER If the loss suddenly becomes NaN, the usual suspects are a learning rate that is too big, a missing gradient clip, or a log of zero somewhere in the loss.

Lesson: Training tricks: schedules, clipping and stopping

STEP 4 Probability and statistics

4.1 Probability and Bayes' Theorem

Probability is the maths of uncertainty. Roll virtual dice, see the addition and multiplication rules on a grid of outcomes, count with permutations and combinations, read conditional probabilities from a table, and sort 500 people to understand Bayes' theorem before building a naive Bayes spam filter.

KEY IDEAS

- **Chance, outcomes and events** Probability is a number from 0 to 1 that says how likely something is. 0 means impossible, 1 means certain, and 0.5 means as likely as not. The list of everything that could happen is the sample space , and any group of outcomes you care about is an event .
- **AND, OR and independence** Two rules combine simple probabilities into bigger ones. OR means "at least one happens". AND means "both happen".
- **Counting: permutations and combinations** When outcomes are equally likely, probability is just counting. Three tools do most of the counting.
- **Conditional probability** A conditional probability answers "how likely is A, now that I know B happened?" Knowing B shrinks the world down to just the outcomes where B is true.
- **Bayes' theorem: updating beliefs** The theorem of Bayes turns P(B | A) around into P(A | B). It is the rule for updating what you believe when new evidence arrives: start with a prior , see evidence, and get a posterior .
- **Naive Bayes: a classifier from counting** Naive Bayes uses the theorem of Bayes to classify, for example spam or not spam. It makes one bold (naive) simplification: given the class, every word is independent of the others.
- **Probability inside machine learning** Most models do not just say "cat". They say "cat with probability 0.92". Probability is how machine learning expresses uncertainty.

FORMULAS

Equally likely outcomes Exam favourite

$$P(A) = \frac{\text{number of outcomes in A}}{\text{total number of outcomes}}$$

A fair die: P(even) = 3/6 = 0.5.

Lesson: Chance, outcomes and events

Complement

$$P(\text{not A}) = 1 - P(A)$$

Often the easy way round: P(at least one six in 4 rolls) = $1 - (5/6)^4 \approx 0.518$.

Lesson: Chance, outcomes and events

Addition rule (OR)

Exam favourite

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Subtract the overlap so it is not counted twice. If A and B cannot happen together (mutually exclusive), the overlap is 0.

Lesson: AND, OR and independence

Multiplication rule (AND)

$$P(A \cap B) = P(A) \times P(B \mid A)$$

If A and B are independent, P(B | A) = P(B) and this becomes P(A) × P(B).

Lesson: AND, OR and independence

Factorial

Exam favourite

$$n! = n \times (n-1) \times \ldots \times 2 \times 1$$

Ways to arrange n different things in a row. 5! = 120. By agreement 0! = 1.

Lesson: Counting: permutations and combinations

Permutations (order matters)

$$P(n, k) = \frac{n!}{(n-k)!}$$

Gold, silver and bronze from 10 runners: 10 × 9 × 8 = 720.

Lesson: Counting: permutations and combinations

Combinations (order does not matter)

$$C(n, k) = \frac{n!}{k!\,(n-k)!}$$

A team of 3 from 10 people: 720 ÷ 3! = 120. Also written ^nC_k or "n choose k".

Lesson: Counting: permutations and combinations

Conditional probability

Exam favourite

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Read the bar | as "given". Out of the B world, what share is also A?

Lesson: Conditional probability

Bayes' theorem

Exam favourite

$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)} \quad \text{with} \quad P(B) = P(B \mid A)P(A) + P(B \mid \text{not } A)P(\text{not } A)$$

posterior = likelihood × prior ÷ evidence.

Lesson: Bayes' theorem: updating beliefs

Bayes in odds form

$$\text{posterior odds} = \text{prior odds} \times \text{likelihood ratio}$$

likelihood ratio = P(evidence | A) ÷ P(evidence | not A). Above 1 the evidence favours A, below 1 it favours not A.

Lesson: Bayes' theorem: updating beliefs

Naive Bayes score

Exam favourite

$$\text{score}(\text{class}) = P(\text{class}) \times \Pi\, P(\text{word}_i \mid \text{class})$$

Compute it for every class and pick the biggest. Dividing by the total turns scores into probabilities.

Lesson: Naive Bayes: a classifier from counting

Laplace (add one) smoothing

$$P(\text{word} \mid \text{class}) = \frac{\text{count} + 1}{\text{total words in class} + \text{vocabulary size}}$$

Without the +1, one unseen word would make the whole product 0.

Lesson: Naive Bayes: a classifier from counting

Monte Carlo estimate

Exam favourite

$$\text{average of f(random samples)} \to \text{true average as samples grow}$$

Throw random darts at a square: the share landing inside the quarter circle tends to π/4.

Lesson: Probability inside machine learning

TABLES

Table: experiment · sample space · event · probability

Experiment	Sample space	Event	Probability
toss a coin	{{H, T}}	heads	1/2
roll a die	{{1, 2, 3, 4, 5, 6}}	more than 4	2/6 = 1/3
roll two dice	36 ordered pairs	sum is 7	6/36 = 1/6
pick a card	52 cards	a king	4/52 = 1/13

Lesson: Chance, outcomes and events

Table: words · meaning · example

Words	Meaning	Example
mutually exclusive	cannot both happen, $P(A \cap B) = 0$	rolling a 2 and rolling a 5 on one die
independent	knowing one tells you nothing about the other	two separate coin tosses
dependent	one changes the chance of the other	drawing two cards without putting the first back

Lesson: AND, OR and independence

Table: · passed · failed · total

	Passed	Failed	Total
Studied	90	10	100
Did not study	40	60	100
Total	130	70	200

Lesson: Conditional probability

Table: version · features · p(feature | class) comes from

Version	Features	P(feature | class) comes from
Multinomial	word counts	counting words in each class, with add one smoothing
Bernoulli	word present or absent	the share of documents in each class that contain the word
Gaussian	numbers such as height or blood pressure	a normal curve with the mean and SD of that feature in that class (Chapter 10)

Lesson: Naive Bayes: a classifier from counting

Table: idea · where it appears

Idea	Where it appears
Predicted probabilities	logistic regression and softmax outputs (Chapter 12)
Likelihood	training often maximises P(data | weights), the maximum likelihood idea of Chapter 11
Prior and posterior	Bayesian models and regularisation, where a prior belief keeps weights small
Sampling	dropout, data augmentation, and how language models pick the next word
Monte Carlo	estimating hard averages by simulating many random cases
Calibration	checking that "90% sure" predictions are right about 90% of the time

Lesson: Probability inside machine learning

TIPS AND TRAPS

REMEMBER In the long run, the share of times an event happens settles near its probability. This is the **law of large numbers**, and the dice animation shows it: with 10 rolls the bars are uneven, with 1000 rolls they are nearly level.
Lesson: Chance, outcomes and events

REMEMBER OR adds (then subtract the overlap). AND multiplies (when independent).
Lesson: AND, OR and independence

COMMON MISTAKE Mutually exclusive events are NOT independent. If A happens, B cannot, so knowing A tells you a lot about B.
Lesson: AND, OR and independence

REMEMBER Ask "does the order matter?" A podium, a PIN and a password are permutations. A team, a hand of cards and a lottery ticket are combinations.
Lesson: Counting: permutations and combinations

REMEMBER When working out C(n, k) by hand, cancel the bigger factorial first and multiply only k numbers on top. C(10, 3) = (10 × 9 × 8) ÷ (3 × 2 × 1) = 120.
Lesson: Counting: permutations and combinations

COMMON MISTAKE P(A | B) and P(B | A) are different numbers. Most people who are fluent in French have been to France, but most people who have been to France are not fluent in French. Mixing them up is called the **prosecutor fallacy**.
Lesson: Conditional probability

REMEMBER Think in whole people, not percentages. Draw 10000 people, split them into sick and healthy, then split each group by test result. The answer is just one count divided by another.
Lesson: Bayes' theorem: updating beliefs

REMEMBER When a probability is too hard to work out on paper, simulate it a million times and count.
Lesson: Probability inside machine learning

4.2 Random Variables and Distributions

A distribution says which values a random quantity is likely to take. Build binomial bars one at a time, shade areas under the bell curve, watch lopsided data turn bell shaped through the central limit theorem, and measure how two variables move together.

KEY IDEAS

- **Random variables, PMF, PDF and CDF** A random variable is a number that comes out of a chance event: the face of a die, the number of heads in ten flips, or the time until the next bus. We write it with a capital letter such as X, and a value it can take with a small letter such as x.
- **Expected value and variance** The expected value E[X] (also called the mean μ) is the long run average of X if you could repeat the experiment forever.
- **Bernoulli and binomial: counting successes** A Bernoulli trial has two results: success (1) with chance p or failure (0) with chance 1 − p. A coin toss, a free throw and "did the user click?" are all Bernoulli trials.
- **Poisson, uniform and exponential** The Poisson distribution counts how many events happen in a fixed stretch of time or space when they arrive at random at an average rate λ (lambda). Calls to a help desk per hour, typos per page and goals per match are classic examples.
- **The normal distribution and z-scores** The normal (or Gaussian) distribution is the famous bell curve. It is set by two numbers: the mean μ, which places the centre, and the standard deviation σ, which sets the width. Heights, measurement errors and exam scores are often close to normal.

- **The central limit theorem** The central limit theorem (CLT) explains why the bell curve is everywhere. Take many independent random values from almost any distribution (it only needs a finite variance), and their average is close to normally distributed.
- **Covariance, correlation and distributions in ML** So far each variable lived alone. Covariance asks whether two variables move together. If X is above its mean when Y is above its mean, the covariance is positive. Correlation rescales covariance to a number between −1 and 1 so it is easy to read.

FORMULAS

PMF (discrete)

Exam favourite

$$p(x) = P(X = x) \quad \text{with} \quad \Sigma\, p(x) = 1$$

Probability mass function: the chance of each exact value. All the bars add up to 1.

Lesson: Random variables, PMF, PDF and CDF

PDF (continuous)

$$P(a \le X \le b) = \int_a^b f(x)\,dx \quad \text{with total area } 1$$

Probability density function: probability is the AREA under the curve between a and b.

Lesson: Random variables, PMF, PDF and CDF

CDF (both)

$$F(x) = P(X \le x)$$

Cumulative distribution function: it climbs from 0 on the far left to 1 on the far right and never goes down.

Lesson: Random variables, PMF, PDF and CDF

Quantile (the inverse of the CDF)

$$\text{the } q \text{ quantile is the smallest } x \text{ with } F(x) \ge q$$

The median is the 0.5 quantile. The 0.95 quantile is the value that 95% of results stay at or below. Weather services and delivery apps quote these: "arrives within 30 minutes 95% of the time".

Lesson: Random variables, PMF, PDF and CDF

Expected value

Exam favourite

$$E[X] = \mu = \Sigma\, x \cdot p(x)$$

Each value times its chance, all added up.

Lesson: Expected value and variance

Variance

$$\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

The average squared distance from the mean. The second form is quicker by hand.

Lesson: Expected value and variance

Standard deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

Same units as X, so it is easier to picture.

Lesson: Expected value and variance

Shifting and scaling

$$E[aX + b] = a\,E[X] + b \qquad \mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$$

Adding b moves the mean but not the spread. Multiplying by a stretches the spread by a.

Lesson: Expected value and variance

Adding variables

$$E[X + Y] = E[X] + E[Y] \qquad \mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

The mean rule is always true. The variance rule needs X and Y to be independent.

Lesson: Expected value and variance

Bernoulli(p)

Exam favourite

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

Mean p, variance p(1 − p).

Lesson: Bernoulli and binomial: counting successes

Binomial(n, p)

$$P(X = k) = C(n, k) \cdot p^k \cdot (1 - p)^{n-k}$$

C(n, k) counts the ways to place k successes among n tries. $p^k(1 - p)^{n-k}$ is the chance of any one of those ways.

Lesson: Bernoulli and binomial: counting successes

Binomial mean and spread

$$\mu = n\,p \qquad \sigma^2 = n\,p\,(1 - p)$$

Ten fair flips: mean 5 heads, variance 2.5, SD about 1.58.

Lesson: Bernoulli and binomial: counting successes

Poisson(λ)

Exam favourite

$$P(X = k) = \frac{e^{-\lambda} \cdot \lambda^{k}}{k!}$$

The mean and the variance are both λ.

Lesson: Poisson, uniform and exponential

Normal PDF

Exam favourite

$$f(x) = \frac{1}{\sigma \sqrt{(2\pi)}} \cdot e^{-(x - \mu)^2 \div (2\sigma^2)}$$

The curve is symmetric about μ and its peak height is $1 \div (\sigma\sqrt{(2\pi)})$.

Lesson: The normal distribution and z-scores

z-score

$$z = \frac{x - \mu}{\sigma}$$

How many standard deviations x is from the mean. Any normal becomes the standard normal N(0, 1) this way.

Lesson: The normal distribution and z-scores

Central limit theorem

Exam favourite

$$\bar{X} = \frac{X_1 + X_2 + \ldots + X_n}{n} \quad \text{is roughly} \quad N(\mu, \sigma^2 \div n)$$

The average has the same centre μ as one value, but a narrower spread.

Lesson: The central limit theorem

Standard error

$$SE = \frac{\sigma}{\sqrt{n}}$$

The SD of an average of n values. It shrinks as the sample grows.

Lesson: The central limit theorem

Sample covariance

Exam favourite

$$Cov(X, Y) = \frac{\Sigma\,(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Positive: they rise together. Negative: one rises as the other falls.

Lesson: Covariance, correlation and distributions in ML

Correlation

$$r = \frac{Cov(X, Y)}{s_x \cdot s_y}$$

Always between −1 and 1. r = 1 is a perfect rising straight line.

Lesson: Covariance, correlation and distributions in ML

TABLES

Table: · discrete · continuous

	Discrete	Continuous
values	a list you can count	any number in a range
described by	PMF: bar heights	PDF: a smooth curve
probability of one exact value	can be above 0	always 0
probability of a range	add the bars	area under the curve
examples	goals in a match, emails per day	height, waiting time, temperature

Lesson: Random variables, PMF, PDF and CDF

Table: distribution · type · parameters · mean

Distribution	Type	Parameters	Mean	Variance	Typical use
Bernoulli	discrete	p	p	$p(1 - p)$	one yes or no
Binomial	discrete	n, p	np	$np(1 - p)$	successes in n tries
Poisson	discrete	λ	λ	λ	events per hour
Uniform	continuous	a, b	$(a + b) \div 2$	$(b - a)^2 \div 12$	a random number in a range
Normal	continuous	μ, σ	μ	σ^2	heights, measurement noise
Exponential	continuous	λ	$1 \div \lambda$	$1 \div \lambda^2$	waiting time until the next event

Lesson: Poisson, uniform and exponential

Table: range · share of values · outside the range

Range	Share of values	Outside the range
$\mu \pm 1\sigma$	68.27%	about 1 in 3
$\mu \pm 2\sigma$	95.45%	about 1 in 22
$\mu \pm 3\sigma$	99.73%	about 1 in 370

Lesson: The normal distribution and z-scores

Table: distribution or idea · where machine learning uses it

Distribution or idea	Where machine learning uses it
Bernoulli	binary labels, and the output of logistic regression (Chapter 12)
Categorical	multi class labels and the softmax output layer
Normal	noise in linear regression, weight initialisation, Gaussian naive Bayes
Poisson	regression on counts such as visits or sales
Central limit theorem	error bars on accuracy, A/B tests (Chapter 11), noise in mini batch gradients
Covariance matrix	PCA (Chapter 5) and correlated features

Lesson: Covariance, correlation and distributions in ML

TIPS AND TRAPS

COMMON MISTAKE For a continuous variable, $P(X = \text{exactly } 1.5) = 0$, because a single point has no width and so no area. Only ranges have probability. That is also why a PDF can be taller than 1: its height is a density, not a probability.

Lesson: Random variables, PMF, PDF and CDF

REMEMBER The expected value does not have to be a possible result. Nobody ever rolls 3.5, yet 3.5 is where the average of many rolls ends up.

Lesson: Expected value and variance

REMEMBER Poisson is what the binomial becomes when n is huge and p is tiny, with $n\,p = \lambda$. Many people (huge n) each with a tiny chance of calling in this hour (tiny p) gives a Poisson number of calls.

Lesson: Poisson, uniform and exponential

REMEMBER Remember 68, 95, 99.7. Within 1, 2 and 3 standard deviations of the mean you find about 68%, 95% and 99.7% of a normal population.

Lesson: The normal distribution and z-scores

REMEMBER The $\sqrt{n}$ rule: 4 times more data halves the error, and 100 times more data cuts it to a tenth. Precision gets expensive.

Lesson: The central limit theorem

REMEMBER Covariance has awkward units (hours × points). Correlation has no units, which is why people quote r instead.

Lesson: Covariance, correlation and distributions in ML

COMMON MISTAKE Correlation is not causation. Ice cream sales and sunburn are correlated because both rise in hot weather. Also, r only measures straight line relationships: for $x = -2, -1, 0, 1, 2$ and $y = x^2$, r is exactly 0 even though y depends completely on x.

Lesson: Covariance, correlation and distributions in ML

4.3 Statistics and Inference

Statistics turns a sample into honest conclusions. Describe data with the right summaries, watch confidence intervals catch (or miss) the true mean, run a hypothesis test and an A/B test step by step, climb a likelihood curve to its peak, and fit a regression line while keeping an eye on overfitting.

KEY IDEAS

- **Describing data: centre, spread and outliers** Statistics turns a pile of numbers into a few that tell the story. The first questions are always: where is the centre, and how spread out is the data?
- **Samples, bias and standard error** We rarely measure everyone. The whole group we care about is the population . The part we actually measure is the sample . A number worked out from the sample (such as $\bar{x}$) is a statistic , and it is our estimate of the population parameter (such as μ).
- **Confidence intervals** A single estimate such as "18.2 hours" hides its uncertainty. A confidence interval gives a range: estimate ± margin of error. A 95% confidence interval is built by a method that catches the true value in 95% of samples.
- **Hypothesis tests and p-values** A hypothesis test asks whether the data is surprising enough to doubt a default claim. The default, "nothing special is going on", is the null hypothesis H_0 . The claim we are testing for is the alternative H_1 .
- **A/B testing** An A/B test is a hypothesis test that companies run every day. Visitors are split at random between the current version A and a new version B, and we compare a rate such as the share who buy.
- **Maximum likelihood estimation** Maximum likelihood answers "which parameter value makes the data we saw most probable?" It is the idea behind how most machine learning models are trained.
- **Regression, R^2 and the bias and variance trade-off** Linear regression fits the straight line $\hat{y} = a + b\,x$ that makes the squared vertical misses (residuals) as small as possible. It is both a statistics tool and the simplest machine learning model.

FORMULAS

Mean Exam favourite

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n}$$

Add everything and share it out equally. Pulled hard by extreme values.

Lesson: Describing data: centre, spread and outliers

Median

the middle value once sorted

With an even count, average the two middle values. Barely moved by extreme values.

Lesson: Describing data: centre, spread and outliers

Mode

the most common value

The only one that works for categories such as favourite colour.

Lesson: Describing data: centre, spread and outliers

Sample variance and SD

$$s^2 = \frac{\Sigma\,(x_i - \bar{x})^2}{n - 1} \qquad s = \sqrt{s^2}$$

Dividing by $n - 1$ instead of n corrects for the fact that data sits closer to its own mean than to the true mean (Bessel correction).

Lesson: Describing data: centre, spread and outliers

Quartiles and IQR

$$\text{IQR} = Q3 - Q1$$

Q1 and Q3 are the medians of the lower and upper halves. The middle 50% of the data lies between them.

Lesson: Describing data: centre, spread and outliers

Outlier fences (Tukey)

$$\text{below } Q1 - 1.5 \times \text{IQR} \text{ or above } Q3 + 1.5 \times \text{IQR}$$

A common rule of thumb for flagging values worth a closer look.

Lesson: Describing data: centre, spread and outliers

Standard error of a mean

Exam favourite

$$\text{SE} = \frac{s}{\sqrt{n}}$$

How much $\bar{x}$ would wobble from sample to sample (Chapter 10).

Lesson: Samples, bias and standard error

Standard error of a proportion

$$\text{SE} = \sqrt{\left(\frac{\hat{p}\,(1-\hat{p})}{n}\right)}$$

540 yes out of 1000: $\sqrt{(0.54 \times 0.46 \div 1000)} \approx 0.0158$.

Lesson: Samples, bias and standard error

Confidence interval for a mean

Exam favourite

$$\bar{x} \pm z^* \cdot \text{SE} \quad \text{with} \quad z^* = 1.645\ (90\%),\ 1.96\ (95\%),\ 2.576\ (99\%)$$

For small samples with unknown σ, statisticians use a slightly wider t value instead of z^*.

Lesson: Confidence intervals

Test statistic (z-test for a mean)

Exam favourite

$$z = \frac{\bar{x} - \mu_0}{\sigma \div \sqrt{n}}$$

How many standard errors the sample mean is from the value the null claims.

Lesson: Hypothesis tests and p-values

p-value

$$p = P(\text{a result at least this extreme} \mid H_0 \text{ is true})$$

Small p means the data would be rare if H_0 were true. Reject H_0 when $p < \alpha$, usually $\alpha = 0.05$.

Lesson: Hypothesis tests and p-values

Pooled rate

Exam favourite

$$\hat{p} = \frac{\text{buys A} + \text{buys B}}{\text{visitors A} + \text{visitors B}}$$

Under H_0 both pages share one true rate, so we pool the data to estimate it.

Lesson: A/B testing

Two proportion z-test

$$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{(\hat{p}(1-\hat{p})(1/n_A + 1/n_B))}}$$

Then the p-value comes from the normal curve, as before.

Lesson: A/B testing

Visitors needed in each group (rule of thumb)

$$n \approx \frac{16 \times p(1-p)}{\delta^2}$$

p is the current rate and δ is the smallest difference worth detecting. The 16 comes from $(1.96 + 0.84)^2 \times 2 \approx 15.7$.

Lesson: A/B testing

Likelihood of coin flips

Exam favourite

$$L(p) = p^h\,(1-p)^{n-h}$$

h heads out of n flips, if the chance of heads is p.

Lesson: Maximum likelihood estimation

Log-likelihood

$$\ell(p) = h \log p + (n-h) \log(1-p)$$

Same best p, but sums instead of products: easier calculus, no underflow.

Lesson: Maximum likelihood estimation

The answer

$$\hat{p} = \frac{h}{n}$$

Set the derivative h/p − (n − h)/(1 − p) to 0 and solve.

Lesson: Maximum likelihood estimation

Least squares line

Exam favourite

$$b = \frac{\Sigma\,(x_i - \bar{x})(y_i - \bar{y})}{\Sigma\,(x_i - \bar{x})^2} \qquad a = \bar{y} - b\,\bar{x}$$

Slope = covariance ÷ variance of x. The line always passes through the point $(\bar{x}, \bar{y})$.

Lesson: Regression, R^2 and the bias and variance trade-off

R^2 (coefficient of determination)

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = 1 - \frac{\Sigma\,(y_i - \hat{y}_i)^2}{\Sigma\,(y_i - \bar{y})^2}$$

The share of the variation in y that the line explains. For a straight line fit, $R^2 = r^2$.

Lesson: Regression, R^2 and the bias and variance trade-off

Bias and variance decomposition

$$\text{expected test error} = \text{bias}^2 + \text{variance} + \text{noise}$$

Bias: errors from a model too simple to capture the pattern. Variance: errors from a model so flexible it chases the random quirks of its training sample.

Lesson: Regression, R^2 and the bias and variance trade-off

TABLES

Table: comparison · shape · everyday example

Comparison	Shape	Everyday example
mean ≈ median	roughly symmetric	heights of adults of one sex
mean > median	long tail to the right (right skew)	incomes, house prices, time spent on a website
mean < median	long tail to the left (left skew)	marks in an easy exam where most score high

Lesson: Describing data: centre, spread and outliers

Table: · population · sample

	Population	Sample
size	N (often huge)	n
mean	μ (unknown)	$\bar{x}$ (we compute it)
SD	σ	s
proportion	p	$\hat{p}$

Lesson: Samples, bias and standard error

Table: kind of bias · what goes wrong · example

Kind of bias	What goes wrong	Example
selection bias	the sample is not like the population	surveying gym members about exercise habits
survivorship bias	only the survivors are visible	studying only successful startups
non-response bias	the people who answer differ from those who do not	a long email survey
measurement bias	the tool itself is off	a scale that always reads 1 kg heavy

Lesson: Samples, bias and standard error

Table: · H_0 really true · H_0 really false

	H_0 really true	H_0 really false
Reject H_0	Type I error (false positive), chance α	correct, chance = power
Keep H_0	correct	Type II error (false negative), chance β

Lesson: Hypothesis tests and p-values

Table: test · question it answers · example

Test	Question it answers	Example
one sample t-test	is the mean different from a claimed value, when σ is unknown	do our batteries last 10 hours on average
two sample t-test	do two groups have different means	is model B faster than model A per request
paired t-test	did the same items change between two conditions	each patient's blood pressure before and after
chi-square test	are two categories related, from a table of counts	is device type related to whether people buy
two proportion z-test	do two rates differ	A/B test of click rates (next lesson)

Lesson: Hypothesis tests and p-values

Table: model · what maximum likelihood gives

Model	What maximum likelihood gives
coin (Bernoulli)	$\hat{p}$ = share of successes
normal data	$\hat{\mu}$ = sample mean, σ^2 = average squared deviation (divides by n)
linear regression with normal noise	minimise the sum of squared errors (least squares)
logistic regression	minimise log loss, also called cross entropy (Chapter 12)

Lesson: Maximum likelihood estimation

Table: · too simple (high bias) · just right · too flexible (high variance)

	Too simple (high bias)	Just right	Too flexible (high variance)
training error	high	low	very low
test error	high	low	high
name	underfitting	good fit	overfitting
fix	more features, bigger model	keep it	more data, regularisation, simpler model

Lesson: Regression, R^2 and the bias and variance trade-off

TIPS AND TRAPS

REMEMBER For skewed data such as incomes or house prices, quote the median. The mean is pulled toward the long tail.

Lesson: Describing data: centre, spread and outliers

COMMON MISTAKE A bigger sample shrinks random error but does nothing about **bias**. Polling only people who answer landline phones, or training a face detector only on photos of one group of people, gives a confident answer to the wrong question.

Lesson: Samples, bias and standard error

REMEMBER To halve the margin of error, you need four times as many people. That √n again.

Lesson: Confidence intervals

COMMON MISTAKE "95% confident" does not mean there is a 95% chance that μ is inside this particular interval. μ is fixed, the interval is what varies. It means that 95% of intervals made this way catch μ. The animation shows exactly that: about 1 in 20 intervals miss.

Lesson: Confidence intervals

REMEMBER If p is low, the null must go. But statistically significant is not the same as important: with a million users, a 0.01% difference can be significant and still not worth anything.

Lesson: Hypothesis tests and p-values

COMMON MISTAKE A p-value is NOT the probability that H_0 is true. And "not significant" does not prove there is no effect. It only says this data was not enough to tell.

Lesson: Hypothesis tests and p-values

REMEMBER Halve the effect you want to see and you need four times the people. The square in δ^2 is the same $\sqrt{n}$ rule seen throughout this chapter.

Lesson: A/B testing

COMMON MISTAKE Decide the sample size before the test and do not stop the moment p dips below 0.05. Peeking every day and stopping at the first "win" makes false positives far more common than 5%.

Lesson: A/B testing

REMEMBER Always judge a model on data it has never seen. A high R^2 on the training data alone says little about the future.

Lesson: Regression, R^2 and the bias and variance trade-off

STEP 5 Maths inside ML models

5.1 Entropy, Loss Functions and Metrics

This is where the maths meets real models. Measure surprise with entropy, see how cross entropy and softmax train classifiers, compare MSE, MAE and log loss, shrink weights with regularisation, and judge a model with precision, recall and the ROC curve.

KEY IDEAS

- **Information and entropy** Some news tells you a lot, some tells you almost nothing. "The sun rose today" is no surprise. "It snowed in the Sahara" is a big surprise. Information theory measures surprise with a number, and entropy is the average surprise of a random event.
- **Cross entropy and KL divergence** Suppose reality follows p, but your model believes q. You will be surprised more often than necessary. Cross entropy measures your average surprise, and the extra over the unavoidable entropy is the KL divergence .
- **Softmax, temperature and log-sum-exp** A classifier first produces raw scores called logits , which can be any real numbers. Softmax turns them into probabilities: all positive and adding to 1, with the biggest logit getting the biggest share.
- **Loss functions for regression** A loss function turns "how wrong was the model?" into one number that training then makes smaller. The choice of loss decides what kind of mistakes the model cares most about.
- **Log loss for classification** For a yes or no classifier that outputs a probability p, the standard loss is log loss , also called binary cross entropy. It is the cross entropy from earlier with only two classes.
- **Regularisation: L2 and L1** A flexible model can fit noise (overfitting, Chapter 11). Regularisation adds a penalty for large weights to the loss, so the model only uses big weights when the data really demands them.
- **Classification metrics, ROC and AUC** Accuracy alone can fool you. If 99% of card payments are honest, a model that never flags fraud is 99% accurate and completely useless. The confusion matrix shows what really happened.

FORMULAS

Surprise (information) of an outcome

Exam favourite

$$I(x) = \log_2 \frac{1}{p(x)} = -\log_2 p(x)$$

Measured in bits. A fair coin flip carries 1 bit. A 1 in 8 event carries 3 bits.

Lesson: Information and entropy

Entropy

$$H(p) = -\Sigma\, p(x) \log_2 p(x)$$

The average surprise. It is 0 when one outcome is certain and largest when every outcome is equally likely.

Lesson: Information and entropy

Mutual information

$$I(X, Y) = H(Y) - H(Y \mid X) = \Sigma\, p(x, y) \log [p(x, y) \div (p(x)\, p(y))]$$

It is 0 exactly when X and Y are independent, and it catches curved and non numeric relationships that correlation misses.

Lesson: Information and entropy

Cross entropy

Exam favourite

$$H(p, q) = -\Sigma\, p(x) \log q(x)$$

Reality p, belief q. It is smallest when q = p.

Lesson: Cross entropy and KL divergence

KL divergence

$$KL(p \parallel q) = \Sigma\, p(x) \log \frac{p(x)}{q(x)} = H(p, q) - H(p)$$

Always ≥ 0, and 0 only when q = p. Not symmetric: $KL(p \parallel q) \neq KL(q \parallel p)$.

Lesson: Cross entropy and KL divergence

Softmax

Exam favourite

$$\text{softmax}(z)_i = \frac{e^{z_i} \div T}{\Sigma_j\, e^{z_j} \div T}$$

T is the temperature. T = 1 is normal, small T sharpens the choice, large T flattens it.

Lesson: Softmax, temperature and log-sum-exp

Stable computation

$$\text{softmax}(z) = \text{softmax}(z - \max z)$$

Subtracting the same number from every logit changes nothing, but keeps e^z from overflowing.

Lesson: Softmax, temperature and log-sum-exp

Log-sum-exp

$$\log \Sigma\, e^{z_j} = m + \log \Sigma\, e^{z_j - m}, \quad m = \max z$$

Cross entropy loss = log-sum-exp(z) − $z_{correct}$.

Lesson: Softmax, temperature and log-sum-exp

Mean squared error

Exam favourite

$$MSE = \frac{1}{n} \Sigma\, (y_i - \hat{y}_i)^2$$

Squares make big misses count a lot. Smooth and easy to differentiate. The best constant prediction is the mean.

Lesson: Loss functions for regression

Huber loss

$$\tfrac{1}{2}\, e^2 \text{ when } |e| \le \delta, \quad \delta(|e| - \tfrac{1}{2}\delta) \text{ otherwise}$$

Squared for small errors, straight line for big ones: the best of both.

Lesson: Loss functions for regression

Binary cross entropy

Exam favourite

$$L = -[\, y \log p + (1 - y) \log(1 - p)\,]$$

If y = 1 only −log p counts. If y = 0 only −log(1 − p) counts.

Lesson: Log loss for classification

L2 (ridge, weight decay)

Exam favourite

$$\text{loss} + \lambda\, \Sigma\, w_j^2$$

Shrinks all weights smoothly toward 0. Rarely makes any weight exactly 0.

Lesson: Regularisation: L2 and L1

L1 (lasso)

$$\text{loss} + \lambda \Sigma |w_j|$$

Pushes some weights to exactly 0, so it also selects features.

Lesson: Regularisation: L2 and L1

One weight ridge solution

$$w = \frac{\Sigma x_i y_i}{\Sigma x_i^2 + \lambda}$$

For $y \approx w\,x$ with no intercept. Bigger λ makes w smaller.

Lesson: Regularisation: L2 and L1

Precision

Exam favourite

$$\frac{TP}{TP + FP}$$

Of everything flagged, how much was right? Important when false alarms are costly.

Lesson: Classification metrics, ROC and AUC

Recall (sensitivity, TPR)

$$\frac{TP}{TP + FN}$$

Of all real positives, how many did we catch? Important when misses are costly.

Lesson: Classification metrics, ROC and AUC

F1 score

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$$

The harmonic mean, high only when both are high.

Lesson: Classification metrics, ROC and AUC

ROC curve and AUC

$$TPR = TP \div (TP + FN) \text{ against } FPR = FP \div (FP + TN) \text{ for every threshold}$$

AUC is the area under that curve: the chance that a random positive gets a higher score than a random negative. 0.5 is guessing, 1 is perfect.

Lesson: Classification metrics, ROC and AUC

F beta score

$$F_\beta = (1 + \beta^2) \times \frac{\text{precision} \times \text{recall}}{\beta^2 \times \text{precision} + \text{recall}}$$

$\beta = 1$ gives F1. $\beta = 2$ counts recall as more important (screening for disease). $\beta = 0.5$ counts precision as more important (spam filtering).

Lesson: Classification metrics, ROC and AUC

TABLES

Table: loss · outliers · gradient · typical use

Loss	Outliers	Gradient	Typical use
MSE	very sensitive	smooth, shrinks near 0	default for regression
MAE	robust	constant size, kink at 0	when outliers are common
Huber	robust	smooth	a safe middle ground

Lesson: Loss functions for regression

Table: error e · mse e² · mae |e| · huber, δ = 1

Error e	MSE e^2	MAE $\lvert e \rvert$	Huber, $\delta = 1$
0.5	0.25	0.5	0.125
1	1	1	0.5
3	9	3	2.5
10 (an outlier)	100	10	9.5

Lesson: Loss functions for regression

Table: true label y · predicted p · loss

True label y	Predicted p	Loss
1	0.9 (confident and right)	0.105
1	0.6 (unsure but right)	0.511
1	0.1 (confident and wrong)	2.303
1	0.01 (very confident and wrong)	4.605

Lesson: Log loss for classification

Table: method · penalty · effect

Method	Penalty	Effect
Ridge (L2)	$\lambda \Sigma w_j^2$	shrinks all weights smoothly, keeps them all
Lasso (L1)	$\lambda \Sigma \lvert w_j \rvert$	sets some weights exactly to 0, choosing features
Elastic net	$\lambda_1 \Sigma \lvert w_j \rvert + \lambda_2 \Sigma w_j^2$	a mix: sparse like L1 but stable when features are correlated

Lesson: Regularisation: L2 and L1

Table: · really positive · really negative

	Really positive	Really negative
Predicted positive	TP (true positive)	FP (false positive)
Predicted negative	FN (false negative)	TN (true negative)

Lesson: Classification metrics, ROC and AUC

TIPS AND TRAPS

REMEMBER Entropy counts the yes or no questions you need on average. A fair die needs $\log_2 6 \approx 2.585$. Finding one person among a million needs about 20.

Lesson: Information and entropy

REMEMBER MSE chases the mean, MAE chases the median. Pick the loss whose "best answer" matches what you want to predict.

Lesson: Loss functions for regression

COMMON MISTAKE Why not use MSE for classification? With a sigmoid output, MSE gives tiny gradients exactly when the model is confidently wrong, so learning stalls. Log loss punishes confident mistakes hard and keeps the gradient $p - y$ large.

Lesson: Log loss for classification

REMEMBER λ is a dial between "trust the data" ($\lambda = 0$) and "keep it simple" (large λ). Pick it with a validation set, never with the test set.

Lesson: Regularisation: L2 and L1

REMEMBER Precision asks "when I shout fraud, am I right?" Recall asks "did I catch all the fraud?" A doctor screening for cancer wants high recall. A spam filter that must not lose real email wants high precision.

Lesson: Classification metrics, ROC and AUC

6.1 Orthogonality, Matrix Decompositions and Tensors

Take linear algebra to the level real libraries use. Build orthonormal bases with Gram Schmidt, factorise matrices with QR, LU and Cholesky, see least squares as a projection, measure how fragile a problem is with the condition number, and master the shape and broadcasting rules of tensors.

KEY IDEAS

- **Orthogonal vectors and orthonormal bases** Two vectors are orthogonal when they meet at a right angle, which happens exactly when their dot product is 0 (Chapter 3). A set of vectors that are all orthogonal to each other and all have length 1 is called orthonormal .
- **Gram Schmidt and the QR decomposition** Most real bases are not orthonormal. The Gram Schmidt process turns any set of independent vectors into an orthonormal set that spans exactly the same space.
- **Quadratic forms and positive definite matrices** A quadratic form is the matrix version of ax^2. For a symmetric matrix A and a vector x, the number x^TAx is a sum of squares and cross terms. In two dimensions it is a surface over the plane, and its shape is decided by A.
- **LU and Cholesky: solving systems fast** Gaussian elimination (Chapter 4) secretly builds a factorisation. Record the multipliers you used to clear each column, put them in a lower triangular matrix L with 1s on the diagonal, and keep the staircase you ended with as an upper triangular matrix U.
- **Least squares as a projection** When there are more equations than unknowns, as with 1000 houses and 3 weights, $X\beta = y$ has no exact answer. Least squares picks the β whose predictions $X\beta$ are as close as possible to y. There is a beautiful picture behind it.
- **Matrix norms, trace and the condition number** Vectors have lengths (Chapter 3). Matrices need a size too, for regularising weight matrices, for measuring how far apart two matrices are, and for judging how sensitive a problem is. There are a few standard choices.
- **Tensors, shapes and broadcasting** A tensor in machine learning code is simply an array with any number of directions, called axes or dimensions. A number is a 0D tensor, a vector is 1D, a matrix is 2D, and so on. The shape lists the size along each axis.

FORMULAS

Orthonormal — Exam favourite

$$q_i \cdot q_j = 1 \text{ if } i = j, \text{ and } 0 \text{ if } i \neq j$$

Each vector has length 1, and different vectors are perpendicular. Put them as columns of a matrix Q and this says $Q^TQ = I$.

Lesson: Orthogonal vectors and orthonormal bases

Coordinates in an orthonormal basis

$$c_i = v \cdot q_i$$

How much of q_i is in v. No equations to solve.

Lesson: Orthogonal vectors and orthonormal bases

Rebuilding the vector

$$v = c_1q_1 + c_2q_2 + \ldots + c_nq_n$$

Add the pieces back together.

Lesson: Orthogonal vectors and orthonormal bases

QR decomposition — Exam favourite

$$A = Q\,R$$

Q: orthonormal columns ($Q^TQ = I$). R: upper triangular, holding the dot products and lengths found during Gram Schmidt.

Lesson: Gram Schmidt and the QR decomposition

Quadratic form in 2D

Exam favourite

$x^T A x$ with $A = [[a, b], [b, c]]$ gives $a\,x_1^2 + 2b\,x_1 x_2 + c\,x_2^2$

It is the curvature part of every smooth function near a flat point, which is why the Hessian (Chapter 7) is a quadratic form.

Lesson: Quadratic forms and positive definite matrices

LU decomposition

Exam favourite

$$A = L\,U$$

L: lower triangular, the multipliers of elimination. U: upper triangular, the staircase. Often rows are swapped first for accuracy, giving PA = LU.

Lesson: LU and Cholesky: solving systems fast

Cholesky decomposition

$$A = L\,L^T$$

Only for symmetric positive definite A. L is lower triangular with positive diagonal. Twice as fast as LU and very stable.

Lesson: LU and Cholesky: solving systems fast

The residual is perpendicular to every column

Exam favourite

$$X^T(y - X\beta) = 0 \implies X^T X\beta = X^T y$$

These are the normal equations. "Normal" is an old word for perpendicular.

Lesson: Least squares as a projection

Pseudo inverse

$\beta = X^+ y$ with $X^+ = (X^T X)^{-1} X^T$ when the columns are independent

For any X, the SVD gives $X^+ = V\Sigma^+ U^T$, which also works when columns are dependent. NumPy calls it pinv.

Lesson: Least squares as a projection

Rule of thumb for lost digits

Exam favourite

$$\text{digits lost} \approx \log_{10} \kappa$$

With $\kappa = 10^8$ and float64 (about 16 digits), only about 8 digits of the answer can be trusted. With float32 (7 digits), none can.

Lesson: Matrix norms, trace and the condition number

TABLES

Table: name · rule for every x ≠ 0 · eigenvalues · shape of $x^T ax$

Name	Rule for every x ≠ 0	Eigenvalues	Shape of $x^T Ax$
positive definite	$x^T Ax > 0$	all positive	a bowl with one lowest point
positive semi definite	$x^T Ax \geq 0$	all ≥ 0	a bowl or a flat bottomed trough
indefinite	positive for some x, negative for others	mixed signs	a saddle
negative definite	$x^T Ax < 0$	all negative	an upside down bowl

Lesson: Quadratic forms and positive definite matrices

Table: measure · formula · meaning

Measure	Formula	Meaning
Frobenius norm	$\|A\|_F = \sqrt{(\Sigma\, a_{ij}^2)}$	treat the matrix as one long vector. Weight decay uses its square
spectral norm	$\|A\|_2 = \sigma max$	the most the matrix can stretch any unit vector
trace	$tr(A) = \Sigma\, a_{ii} = \Sigma\, \lambda_i$	sum of the diagonal, also the sum of the eigenvalues
condition number	$\kappa(A) = \sigma max \div \sigma min$	how much small input errors can be amplified

Lesson: Matrix norms, trace and the condition number

Table: data · typical shape · axes read left to right

Data	Typical shape	Axes read left to right
one number (a loss)	()	nothing
a batch of feature vectors	(32, 20)	examples, features
a batch of colour images (PyTorch order)	(32, 3, 224, 224)	examples, colour channels, height, width
a batch of sentences after embedding	(16, 128, 768)	sentences, tokens, embedding numbers
attention scores in a transformer	(16, 12, 128, 128)	sentences, heads, query token, key token

Lesson: Tensors, shapes and broadcasting

TIPS AND TRAPS

REMEMBER Gram Schmidt in four words: project, subtract, normalise, repeat.

Lesson: Gram Schmidt and the QR decomposition

REMEMBER Read shapes from the right, and say "equal or one" for each pair. If any pair fails that test, the operation is an error.

Lesson: Tensors, shapes and broadcasting

6.2 Taylor Series, Newton's Method and Automatic Differentiation

See the calculus that runs inside training code. Approximate functions with Taylor polynomials, solve and minimise with Newton's method, use matrix calculus rules with a shape check, watch automatic differentiation run forward and backward, understand vanishing and exploding gradients, use convexity and Jensen's inequality, and keep numbers stable on a real computer.

KEY IDEAS

- **Taylor series: curves as polynomials** Computers can only add and multiply. To find e 0.5 or sin 0.3, a calculator uses a polynomial that copies the function very closely near one point. That copy is a Taylor polynomial .
- **Newton's method for roots and minimums** To solve f(x) = 0, Newton's method draws the tangent line at your current guess and jumps to the point where that line crosses zero. The tangent line is the order 1 Taylor polynomial, so each step solves a simpler, straight line version of the problem.
- **Matrix calculus: gradients without index pain** When a function takes a whole vector or matrix as input, we want its gradient as a vector or matrix of the same shape.
- **Automatic differentiation: forward and reverse mode** There are three ways a computer can find derivatives. Numerical differences, (f(x + h) − f(x)) ÷ h, are easy but slow and slightly inaccurate. Symbolic differentiation, like a computer algebra system, produces exact formulas that can grow huge.
- **Vanishing and exploding gradients** Backpropagation through a deep network multiplies many factors together, one per layer. Each factor is roughly (slope of the activation) × (weight). If those factors are a little below 1, their product shrinks towards 0 and the early layers stop learning.
- **Convexity and Jensen's inequality** A function is convex when the straight line (chord) between any two points on its graph never goes below the graph. Think of the shape of a bowl.
- **Numerical stability: when maths meets floating point** Floating point numbers (Chapter 1) have about 16 significant digits in float64 and about 7 in float32. Formulas that are perfect on paper can lose all their accuracy on a computer. Three situations come up again and again in machine learning.

FORMULAS

Taylor polynomial around 0 (also called a Maclaurin polynomial)

Exam favourite

$$f(x) \approx f(0) + f'(0)\,x + \frac{f''(0)}{2!}\,x^2 + \frac{f'''(0)}{3!}\,x^3 + \dots$$

n! means 1 × 2 × … × n. Around another point a, replace x by (x − a) and the derivatives at 0 by the derivatives at a.

Lesson: Taylor series: curves as polynomials

Newton step for a root

Exam favourite

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Divide the height by the slope to find how far to slide along the tangent.

Lesson: Newton's method for roots and minimums

Newton step for a minimum

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

A minimum is a root of the slope f', so use the same rule on f'. In many dimensions: $x \leftarrow x - H^{-1}\nabla f$.

Lesson: Newton's method for roots and minimums

Gradient reaching layer 1 of an L layer chain

Exam favourite

$$\partial L/\partial h_1 = \partial L/\partial h_L \times \prod (\sigma'(z_k)\, w_k)$$

One factor per layer. $0.9^{20} \approx 0.12$, $0.5^{20} \approx 0.000001$, $1.5^{20} \approx 3325$.

Lesson: Vanishing and exploding gradients

Definition

Exam favourite

$f(t\,a + (1 - t)\,b) \le t\,f(a) + (1 - t)\,f(b)$ for all t in [0, 1]

The curve at the mixed point is at most the mix of the curve values.

Lesson: Convexity and Jensen's inequality

Quick tests

$f''(x) \ge 0$, or in many dimensions a positive semi definite Hessian

x^2, e^x, $|x|$ and $-\ln x$ are convex. ln x and √x are concave (the chord lies below).

Lesson: Convexity and Jensen's inequality

Jensen's inequality

$f(E[X]) \le E[f(X)]$ for convex f

The function of the average is at most the average of the function. For concave f the sign flips.

Lesson: Convexity and Jensen's inequality

The log-sum-exp trick

Exam favourite

$$\ln \Sigma\, e^{z_i} = m + \ln \Sigma\, e^{z_i - m} \text{ with } m = \max z_i$$

Every exponent is now 0 or below, so nothing overflows, and the biggest term is exactly $e^0 = 1$, so nothing underflows to a sum of 0.

Lesson: Numerical stability: when maths meets floating point

Forward difference

error ≈ h + ε ÷ h, best $h \approx \sqrt{\varepsilon} \approx 10^{-8}$

Gives about 8 correct digits at best.

Lesson: Numerical stability: when maths meets floating point

Central difference

$(f(x + h) - f(x - h)) \div 2h$, error ≈ $h^2 + \varepsilon \div h$, best $h \approx 10^{-5}$

Gives about 10 to 11 correct digits.

Lesson: Numerical stability: when maths meets floating point

TABLES

Table: function · series around 0 · works for

Function	Series around 0	Works for
e^x	$1 + x + x^2/2! + x^3/3! + x^4/4! + \ldots$	every x
sin x	$x - x^3/3! + x^5/5! - x^7/7! + \ldots$	every x
cos x	$1 - x^2/2! + x^4/4! - x^6/6! + \ldots$	every x
ln(1 + x)	$x - x^2/2 + x^3/3 - x^4/4 + \ldots$	$-1 < x \leq 1$ only
1 ÷ (1 − x)	$1 + x + x^2 + x^3 + \ldots$	$-1 < x < 1$ only

Lesson: Taylor series: curves as polynomials

Table: function of x (a vector) · gradient ∇_x · single number version

Function of x (a vector)	Gradient ∇_x	Single number version
a^Tx	a	(ax)' = a
$x^Tx = \|x\|^2$	2x	$(x^2)' = 2x$
x^TAx	$(A + A^T)x$, which is 2Ax when A is symmetric	$(ax^2)' = 2ax$
$½(x - m)^TA(x - m)$, A symmetric	A(x − m)	$(½a(x - m)^2)' = a(x - m)$
$\|Xw - y\|^2$ (as a function of w)	$2X^T(Xw - y)$	$((xw - y)^2)' = 2x(xw - y)$
Wx (vector output)	Jacobian is W	(wx)' = w

Lesson: Matrix calculus: gradients without index pain

Table: · forward mode · reverse mode

	Forward mode	Reverse mode
direction	inputs to output, together with the values	values first, then output back to inputs
carries	the derivative of every node with respect to one input	the derivative of the output with respect to every node
cost	one pass per input	one backward pass per output
best when	few inputs, many outputs	many inputs, one output (a loss)
also called	dual numbers, Jacobian vector product	backpropagation, vector Jacobian product

Lesson: Automatic differentiation: forward and reverse mode

Table: fix · what it does

Fix	What it does
ReLU and its relatives	slope exactly 1 for positive inputs, so the activation does not shrink the signal
careful initialisation	Xavier (variance 1 ÷ n_in) or He (variance 2 ÷ n_in) keeps the size of signals about the same from layer to layer
batch or layer normalisation	rescales the values inside the network so they stay in a healthy range
residual connections	h + F(h) adds a "+1" path, so the gradient can skip layers unchanged
gradient clipping	caps the gradient length to stop explosions, common in recurrent networks
LSTM and GRU gates	let a recurrent network keep a memory path whose factor stays near 1

Lesson: Vanishing and exploding gradients

TIPS AND TRAPS

REMEMBER

Newton in one sentence: replace the curve by its tangent (or the bowl by its quadratic), solve that easy problem, repeat.

Lesson: Newton's method for roots and minimums

REMEMBER Many numbers multiplied together: slightly below 1 vanishes, slightly above 1 explodes. Keep each factor near 1.

Lesson: Vanishing and exploding gradients

REMEMBER Never compute e^big directly. Subtract the max first. Never multiply many probabilities. Add their logs.

Lesson: Numerical stability: when maths meets floating point

6.3 Bayesian Inference, Markov Chains and Sampling

Handle uncertainty the way modern models do. Read joint, marginal and conditional distributions, measure distance with the multivariate normal, update beliefs with Bayes and conjugate priors, watch Markov chains settle, estimate with Monte Carlo, sample posteriors with Metropolis, and see how concentration inequalities guarantee how close an average is.

KEY IDEAS

- **Joint, marginal and conditional distributions** Real data has several variables at once: the weather and whether people carry umbrellas, a patient's age and blood pressure, a pixel and its neighbour. The joint distribution P(X, Y) gives the chance of every combination. From it you can get everything else.
- **The multivariate normal and Mahalanobis distance** The bell curve of Chapter 10 has a version for many variables at once, the multivariate normal . It is described by a mean vector μ (the centre) and a covariance matrix Σ (the spread of each variable and how pairs move together).
- **Bayesian inference: priors, posteriors and MAP** Chapter 9 used Bayes' rule for events. Bayesian inference uses it for an unknown parameter , such as the chance θ that a coin lands heads. Instead of one best guess, you keep a whole distribution of belief over θ and update it with data.
- **Markov chains and stationary distributions** A Markov chain moves between states at random, and the chance of the next state depends only on the current one, not on the history. Weather from day to day, a shopper moving between web pages and the next word in a very simple text model are common examples.
- **Monte Carlo and inverse transform sampling** Monte Carlo methods answer questions by simulation. To find an average, a probability or an integral, draw many random samples and average the results.
- **MCMC and the Metropolis algorithm** Bayesian posteriors are often known only up to a constant: we can compute likelihood × prior, but not p(data), which needs an impossible integral. Markov chain Monte Carlo (MCMC) still produces samples.
- **Concentration inequalities and the law of large numbers** The law of large numbers says that the average of many independent samples gets close to the true mean. Concentration inequalities say how close , with a guarantee rather than an approximation.

FORMULAS

Marginal (sum out the other variable)

Exam favourite

$$P(X = x) = \Sigma_y P(X = x, Y = y)$$

Add across a row of the joint table. The name comes from writing these totals in the margin.

Lesson: Joint, marginal and conditional distributions

Conditional (zoom into one row or column)

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

Keep only the cases where X = x, then rescale so they add to 1.

Lesson: Joint, marginal and conditional distributions

Independence

$$P(X, Y) = P(X)\,P(Y) \text{ for every cell}$$

Knowing X tells you nothing about Y. Then P(Y | X) = P(Y).

Lesson: Joint, marginal and conditional distributions

Density of a d dimensional normal

Exam favourite

$$p(x) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}}\,\exp(-\tfrac{1}{2}\,(x-\mu)^{T}\,\Sigma^{-1}\,(x-\mu))$$

The part inside exp is minus half the squared Mahalanobis distance. |Σ| is the determinant.

Lesson: The multivariate normal and Mahalanobis distance

Mahalanobis distance

$$d_M(x) = \sqrt{((x-\mu)^{T}\,\Sigma^{-1}\,(x-\mu))}$$

Distance measured in standard deviations, taking the shape of the data into account. With Σ = I it is the ordinary distance.

Lesson: The multivariate normal and Mahalanobis distance

Bayes' rule for a parameter

Exam favourite

$$p(\theta \mid \text{data}) = \frac{p(\text{data} \mid \theta)\cdot p(\theta)}{p(\text{data})} \propto \underbrace{\text{likelihood} \times \text{prior}}$$

Prior: belief before the data. Likelihood: how well each θ explains the data. Posterior: updated belief. p(data) only rescales.

Lesson: Bayesian inference: priors, posteriors and MAP

Beta Binomial update

$$\text{Beta}(a, b) \text{ prior} + h \text{ heads}, t \text{ tails} \rightarrow \text{Beta}(a + h, b + t) \text{ posterior}$$

Updating is just counting. The posterior mean is (a + h) ÷ (a + b + h + t).

Lesson: Bayesian inference: priors, posteriors and MAP

Transition matrix

Exam favourite

$$P_{ij} = P(\text{next} = j \mid \text{now} = i), \text{ each row adds up to } 1$$

Row i lists where you can go from state i.

Lesson: Markov chains and stationary distributions

One step for a whole distribution

$$\pi_{t+1} = \pi_t\,P$$

A row vector of probabilities times the matrix.

Lesson: Markov chains and stationary distributions

Stationary distribution

$$\pi = \pi\,P$$

Once reached, the distribution never changes. It is a left eigenvector of P with eigenvalue 1 (Chapter 5).

Lesson: Markov chains and stationary distributions

Monte Carlo estimate

Exam favourite

$$E[f(X)] \approx \frac{1}{n}\,\Sigma\,f(x_i), \text{ with } x_i \text{ drawn from } X$$

An integral $\int_0^1 f(x)\,dx$ is E[f(U)] for a uniform U.

Lesson: Monte Carlo and inverse transform sampling

Its error

$$\text{standard error} = \sigma \div \sqrt{n}$$

Four times as many samples halve the error. This holds in any number of dimensions, which is why Monte Carlo wins in high dimensions.

Lesson: Monte Carlo and inverse transform sampling

Inverse transform

$$x = F^{-1}(u),\ u \sim \text{Uniform}(0, 1)$$

Exponential with rate λ: $F(x) = 1 - e^{-\lambda x}$, so $x = -\ln(1 - u) \div \lambda$.

Lesson: Monte Carlo and inverse transform sampling

TABLES

Table: estimate · rule · beta(2, 2) prior, 7 heads in 10

Estimate	Rule	Beta(2, 2) prior, 7 heads in 10
MLE (maximum likelihood)	the θ that makes the data most likely: h ÷ n	0.700
MAP (maximum a posteriori)	the peak of the posterior: (a + h − 1) ÷ (a + b + n − 2)	8 ÷ 12 = 0.667
posterior mean	(a + h) ÷ (a + b + n)	9 ÷ 14 = 0.643

Lesson: Bayesian inference: priors, posteriors and MAP

Table: step size · what happens · fix

Step size	What happens	Fix
too small	almost every move is accepted, but the chain creeps and samples are strongly correlated	larger steps
too large	most proposals land in low probability areas and are rejected, so the chain sits still	smaller steps
about right	acceptance roughly 25 % to 50 %, and the chain explores well	tune during burn in

Lesson: MCMC and the Metropolis algorithm

Table: inequality · statement · needs

Inequality	Statement	Needs
Markov	$P(X \geq a) \leq E[X] \div a$	X never negative
Chebyshev	$P(\lvert X - \mu \rvert \geq k\sigma) \leq 1 \div k^2$	a finite variance
Chebyshev for an average	$P(\lvert \bar{x} - \mu \rvert \geq \varepsilon) \leq \sigma^2 \div (n\,\varepsilon^2)$	independent samples
Hoeffding	$P(\lvert \bar{x} - \mu \rvert \geq \varepsilon) \leq 2\,e^{-2n\varepsilon^2}$	independent samples between 0 and 1

Lesson: Concentration inequalities and the law of large numbers

TIPS AND TRAPS

REMEMBER

Metropolis in five words: propose, compare, accept or stay.

Lesson: MCMC and the Metropolis algorithm